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Rejuvenation-Responsive and Senolytic-Sensitive Muscle Stem Cells Unveiled by CD200 and CD63 in Geriatric Muscle

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eLife Assessment

This study provides **valuable** insights into aged muscle stem cell biology by revealing phenotypic and functional heterogeneity within the geriatric MuSC pool and proposing a VCam-low/negative subpopulation that may account for the reported decline in MuSC numbers with age. These findings have implications for understanding aging-related changes in stem cell maintenance and for improving strategies to isolate or rejuvenate aged MuSCs. However, the evidence supporting the main claims is **incomplete**, key analyses such as absolute MuSC quantification, fate assessment of VCam-low/negative cells, inclusion of standard aged cohorts, and validation of proposed surface markers are still needed to confirm that overall MuSC abundance is maintained and that a distinct subpopulation has been identified.

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Abstract

Muscle stem cells (MuSCs) are parenchymal cells in skeletal muscle regeneration and maintenance. With aging, MuSCs experience a decline in their regenerative function and reduction in their number. However, recent evidence points to substantial heterogeneity within the aged MuSC population, raising questions about the underlying mechanisms of age-associated dysfunction. Here, we used Pax7^{CreERT2};Rosa^{YFP} mice (MuSC^{YFP}) to label Pax7-expressing MuSCs and chronologically traced MuSCs until geriatric age. Genetic labeling and chronological tracing revealed that the number of YFP⁺ MuSC remained comparable between young, middle and geriatric ages. At geriatric age, YFP⁺ MuSCs exhibited reduced expression of traditional MuSC markers such as VCAM1 and PAX7. A previously unrecognized subpopulation emerged, characterized by loss of VCAM1 and low or absent PAX7. Despite their altered marker profile, these cells retained transcriptional signatures of quiescence and myogenic potential, but displayed significantly reduced proliferative and regenerative capacities. They displayed gene expression patterns indicative of senescence-like state and were selectively ablated by senolytic treatment. DHT restored regenerative function in aged mice and re-induced VCAM1 expression in YFP⁺/Pax7^{low}/VCAM1⁻ cells, indicating responsiveness to rejuvenation. Based on their emergence with aging, functional impairment and responsiveness to rejuvenation, we termed this population GERI-MuSCs (Geriatric Emerging Rejuvenation-responsive and Impaired MuSCs). CD63 and CD200 were identified as novel surface markers that together with VCAM1, reliably detect GERI-MuSCs as well as classical Pax7⁺/VCAM1^{High} MuSCs, providing a tool for comprehensive isolation of MuSCs

from aged wild-type mice. Together, our findings provide a refined framework for studying MuSC aging and offer new tools for isolating functionally distinct MuSC subsets from aged skeletal muscle.

Introduction

Skeletal muscle regeneration is orchestrated primarily by tissue resident stem cells, known as muscle stem cells (MuSCs) or satellite cells (Mauro, 1961 [\[1\]](#)). MuSCs reside beneath the basal lamina of myofibers in a quiescent state under homeostatic condition but become rapidly activated upon injury to proliferate, differentiate, and fuse into regenerating myofibers (Evano and Tajbakhsh, 2018 [\[2\]](#)). While the regenerative function of MuSCs is well maintained in youth, a decline in their number and regenerative competence is observed in aging, leading to impaired muscle repair and tissue maintenance (Blau et al., 2015 [\[3\]](#)).

The age-related decline in MuSC function arises from a combination of extrinsic and intrinsic factors. Extrinsic factors include the increased pro-inflammatory cytokines (Degens, 2007 [\[4\]](#)), extracellular matrix remodeling (Schuler et al., 2021 [\[5\]](#)) and altered mechanical properties of the niche microenvironment (Chakkalakal et al., 2012 [\[6\]](#)). MuSC intrinsic factors include disrupted metabolic homeostasis (Pala et al., 2018 [\[7\]](#)), loss of proteostasis (Kim et al., 2021b [\[8\]](#)), altered epigenetic regulation (Hernando-Herraez et al., 2019 [\[9\]](#)), mitochondrial dysfunction (Rygiel et al., 2016 [\[10\]](#)) and aberrant activation of signaling pathways, including p38 MAPK (Bernet et al., 2014 [\[11\]](#)), JAK/STAT (Price et al., 2014 [\[12\]](#)) and mTORC1 (Tang et al., 2019 [\[13\]](#)). As a result, aged MuSCs show reduced colony-forming capacity, failure to maintain quiescence and loss of self-renewal potential causing a significant decline in the MuSC pool. This reduction in the resident stem cell population directly undermines the muscle's capacity for effective repair (Blau et al., 2015 [\[3\]](#), Day et al., 2010 [\[14\]](#)). Although functional and molecular differences between young and aged MuSCs have been extensively documented, the timing and trajectory by which these alterations arise remain unclear. This study was therefore designed to delineate the transition that occurs between middle-aged and geriatric MuSCs, rather than to re-characterize differences that are already well established between young and aged muscles.

Efforts to reverse MuSC dysfunction have explored both systemic and cell-intrinsic interventions. Circulating factors identified in parabiosis experiments (Conboy et al., 2005 [\[15\]](#)), as well as supplementation with molecules such as NAD⁺ (Zhang et al., 2016 [\[16\]](#)) and oxytocin (Elabd et al., 2014 [\[17\]](#)), have shown potential in restoring regenerative capacity. Pharmacological modulation of signaling pathways, such as inhibition of JAK/STAT (Price et al., 2014 [\[12\]](#)), p38 MAPK (Bernet et al., 2014 [\[11\]](#)), mTORC1, TGF- β or Wnt signaling (Carlson et al., 2009 [\[18\]](#)), has led to improved regeneration in aged muscle. However, variation in the age of experimental animals complicates the interpretation of these findings. Many rejuvenation studies define 'aged' mice as 18 to 24 months old. This range encompasses a period of heterogeneous functional decline, during which a substantial fraction of MuSCs may still retain regenerative capacity. Therefore, since it is unclear whether these models fully represent a truly geriatric state, questions remain as to whether these interventions directly restore dysfunctional MuSCs, selectively expand a subset of aged MuSCs that retain partial functionality, or delay or prevent the aging of MuSCs that have not yet undergone substantial aging. To address this uncertainty, we chronologically traced the fate of Pax7-expressing MuSCs in geriatric mice using a genetic labeling system. This allowed us to evaluate the composition, functionality, and responsiveness of aged MuSC subsets.

In addition to interventions aimed at reversing MuSC dysfunction, an alternative strategy to restore regenerative competence involves targeting senescent cells. Senescent cells accumulate in aged muscle and are thought to impair regeneration through secretion of pro-inflammatory and matrix-remodeling factors, collectively known as the senescence-associated secretory phenotype (SASP) (Saito and Chikenji, 2021 [\[19\]](#)). Senolytic agents, such as dasatinib and quercetin, which selectively eliminate senescent cells have been shown to improve tissue homeostasis and extend muscle healthspan in preclinical models (Xu et al., 2018 [\[20\]](#)). However, although senescence is broadly defined as a state of irreversible cell cycle arrest accompanied by transcriptional reprogramming, how this process manifests in quiescent adult stem cells *in vivo* remains

incompletely understood. This uncertainty raises concerns regarding the use of senolytics, as such treatment may inadvertently eliminate MuSC subsets that retain latent regenerative potential. A better understanding of how senescence-like features emerge within the aged MuSC pool is therefore critical for evaluating the risks and benefits of senolytic treatment in aging muscle.

In this study, our findings reveal an age-dependent accumulation of a previously unrecognized population of MuSCs, which we termed GERI-MuSCs (Geriatric Emerging Regeneration-capable and Impaired MuSCs). GERI-MuSCs are characterized by low expression of traditional marker genes, impaired yet retained functionality, susceptibility to senolytic deletion, and responsiveness to DHT rejuvenation. Moreover, CD63 and CD200 can be reliable markers that can be used with VCAM1 to detect previously hard-to-detect MuSCs. This combinatorial strategy provides a framework for future isolation and therapeutic targeting of GERI-MuSCs. Together, our results refine the understanding of *in vivo* MuSC aging and highlight the importance of studying stem cell subsets that are functionally impaired yet not irreversibly senescent.

Results

Genetic labeling of Pax7⁺ cells reveal marker-deficient MuSCs in aged muscle

Paired box protein 7 (Pax7) is a transcription factor widely recognized as a defining marker of MuSCs (Seale et al., 2000 [DOI](#)). To investigate how aging alters the identity of MuSCs, we used *Pax7^{CreERT2};Rosa^{YFP}* mice (MuSC^{YFP}). In this model, tamoxifen administration at 3-5 months of age irreversibly labels Pax7-expressing MuSCs. YFP⁺ cells were isolated at young (3-5 months), middle (10-14 months) and geriatric (>28 months) ages, using a conventional surface markers and gating strategy: stem cell antigen-1 negative (SCA-1⁻) and vascular cell adhesion protein 1 positive (VCAM1⁺) cells among lineage negative (CD31⁻/CD45⁻) and live cells (Liu et al., 2015 [DOI](#)) (Fig. 1A [DOI](#)). As previously reported, the number of VCAM1⁺/SCA-1⁻ MuSCs declined significantly with age, from an average of 12.22% at young and 13.62% at middle age to 8.31% at geriatric age (Fig. 1B [DOI](#)). The variability of VCAM1⁺ MuSC number at geriatric age likely reflects mouse-to-mouse differences in the onset and progression of late-life phenotypes. Despite this heterogeneity, the overall shift toward a VCAM1-low state was observed across all geriatric replicates. Interestingly, however, when MuSCs were defined by YFP expression alone, their number remained comparable across ages (Fig. 1C [DOI](#)). The broader spread observed in the young group likely reflects the known heterogeneity of MuSCs in early adulthood, which encompass multiple transcriptional and functional substates (Dong Seong Cho, 2017 [DOI](#)). This developmental heterogeneity results in greater variability in young mice compared to the more uniform profiles seen in middle-aged. Although the young group displayed greater biological variability, the overall mean MuSC frequencies were comparable among age groups.

To investigate the disparity in the numbers of surface marker labelled MuSCs and genetically labelled MuSCs at geriatric age, we examined the expression of traditional MuSC surface markers, VCAM1 and integrin alpha 7 (ITGA7) in YFP⁺ population (Blanco-Bose et al., 2001 [DOI](#), Sacco et al., 2008 [DOI](#)). We observed a significant decline in VCAM1 expression, while ITGA7 expression remained comparable (Fig. 1D [DOI](#)). Notably, both VCAM1 and ITGA7 expression showed significantly greater variation at geriatric age, in YFP⁺ cells, likely due to inter-individual variability during aging. In young and middle-aged muscle, the VCAM1^{low} and VCAM1⁻ MuSC populations were extremely small and often nearly absent, indicating that these subsets may be not a substantial component of the MuSC pool earlier in life. The detectable emergence and expansion of VCAM1^{low} cells occurred predominantly at the geriatric stage, suggesting that this population represents an aging-associated state rather than a pre-existing subset.

Since YFP⁺ cells at geriatric age have varying marker gene expression, we investigated their localization *in vivo* by conducting immunohistochemistry of *tibialis anterior* (TA) muscle sections for PAX7. YFP⁺ cells showed sublamellar localization and co-expressed PAX7 in middle aged mice

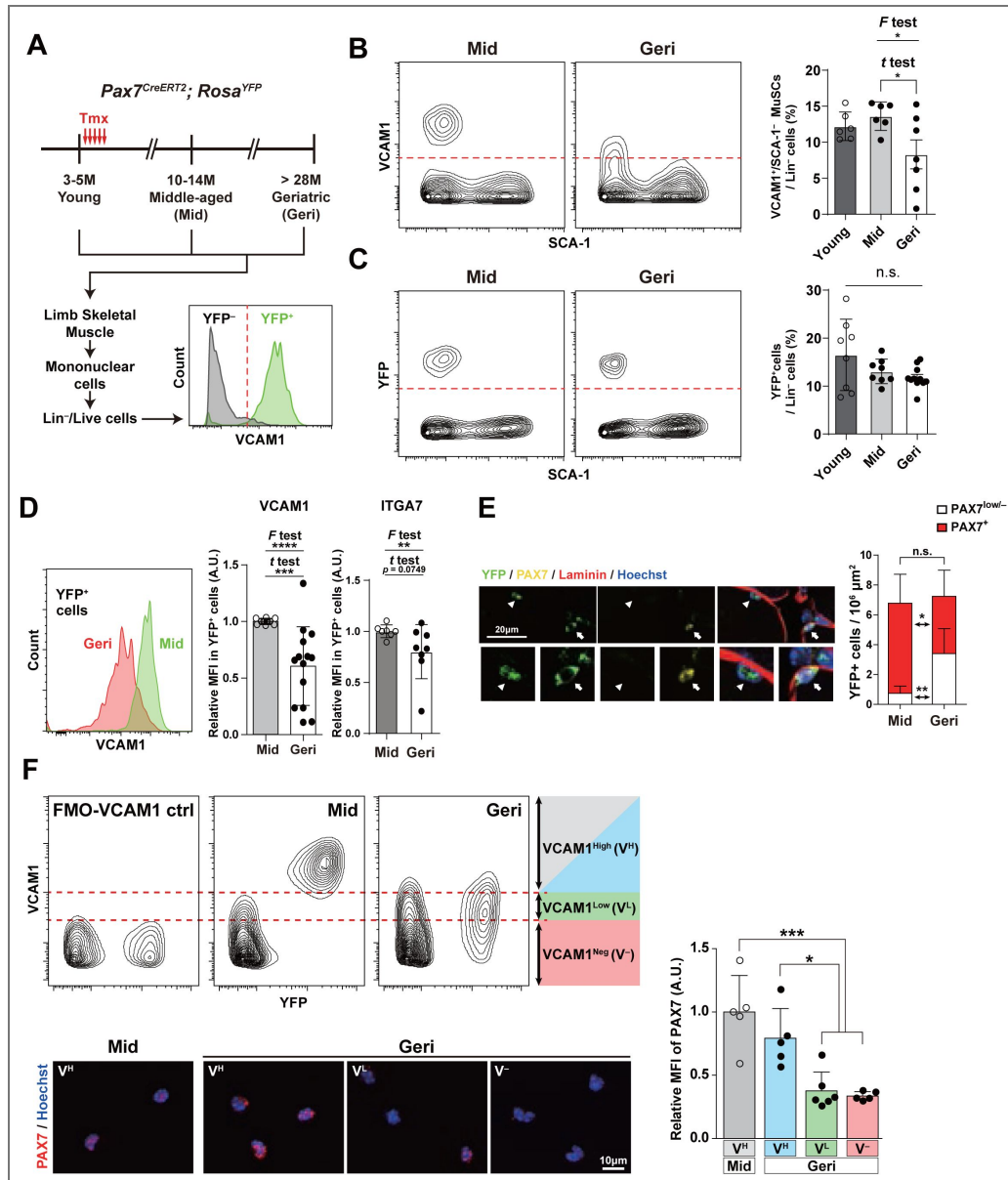


Figure 1. The YFP⁺ cells from *Pax7^{CreERT2}; Rosa^{YFP}* mice do not decrease with age.

(A) Scheme of MuSC labeling in *Pax7^{CreERT2}; Rosa^{YFP}* mice (MuSC^{YFP}) and analysis at middle (Mid) and geriatric age (Geri) using fluorescence-activated cell sorting (FACS) strategy for isolation of YFP⁺ MuSCs using traditional marker, VCAM1. (B, C) Representative plot and quantification for VCAM1⁺ MuSCs (B) and YFP⁺ cells (C) in lineage negative (Lin⁻) cells of Young (n=6 and n=8 each), Mid (n=7, n=8, each) and Geri (n=6, n=11, each) MuSC^{YFP}, Welch's *t* test and *F*-test; *p* = 0.0441 and 0.0460 for (B), respectively. (D) Representative FACS histogram and quantification of VCAM1 and ITGA7 expression in YFP⁺ cells of Mid and Geri MuSC^{YFP}. Quantification of relative VCAM1 and ITGA7 mean fluorescence intensity (MFI) in YFP⁺ cells of Mid (n=12, n=6, each) and Geri (n=14, n=8, each) MuSC^{YFP}, Welch's *t* test and *F*-test; *p* = 0.0009 and <0.0001. (E) Representative images and quantification of immunohistochemistry staining for Hoechst (Blue), YFP (Green), PAX7 (Yellow), and Laminin (Red) in the *Tibialis Anterior* (TA) of MuSC^{YFP}. Arrows indicate PAX7⁺/YFP⁺ cells and arrowheads indicate PAX7⁺/YFP⁺ cells under basal lamina. Quantification of the number of Pax7⁺/YFP⁺ and PAX7⁻/YFP⁺ cells from IHC staining of TA in Mid (n=7) and Geri (n=9) MuSC^{YFP}, two-way ANOVA followed by Bonferroni's multiple comparison test, PAX7⁺/YFP⁺; *p* = 0.0205, PAX7⁻/YFP⁺; *p* = 0.0051. (F) Division of YFP⁺ cells according to their VCAM1 MFI as VCAM1^{High} (V^H), VCAM1^{Low} (V^L), and VCAM1^{Neg} (V⁻) followed by representative images and quantification of immunocytochemistry (ICC) staining for Hoechst (Blue), PAX7 (Red) of YFP⁺ cells. VCAM1 low and negative MuSCs represent approximately 44.6% ± 35.7%, whereas VCAM1 high MuSCs represent 3.9% ± 1.8%. Quantification of PAX7 fluorescence intensity from isolated YFP⁺ cells of Mid V^H, Geri V^H, V^L (n=5, each), and V⁻ (n=6), one-way ANOVA followed by Bonferroni's multiple comparison test; *p* = 0.0004 (vs Mid V^H), and 0.0179, 0.0123 (Geri V^H vs V^L, V⁻, respectively).

(Fig. 1E). In geriatric aged mice, however, the number of PAX7⁺/YFP⁺ cells decreased, while YFP⁺/PAX7^{low/-} cells concomitantly increased (Fig. 1E).

To examine the relationship between PAX7 and varying MuSC marker expression, YFP⁺ cells were grouped by VCAM1 expression levels: VCAM1^{High}, VCAM1^{Low}, and VCAM1⁻, and immunostained for PAX7. The PAX7 fluorescence intensity was comparable between VCAM1^{High} cells from middle-aged and geriatric mice, but was markedly reduced in VCAM1^{Low} and VCAM1⁻ cells of geriatric mice (Fig. 1F). Together, these findings show an emerging population of YFP⁺ cells that lose traditional markers like Pax7 and VCAM1 at geriatric age.

YFP⁺/VCAM1⁻ cells are quiescent MuSCs

To further explore the characteristics of YFP⁺/VCAM1⁻ cells in geriatric mice, we examined their potential myogenic status. Given that loss of canonical markers might reflect lineage progression, we assessed the expression of myogenin (MYOG), a marker of late-stage myogenic differentiation, in YFP⁺ cells from uninjured middle, geriatric ages and 5-days post injury. We found that MYOG expression was largely absent in YFP⁺ cells from uninjured aged muscle, even in geriatric YFP⁺/VCAM1^{Low} and YFP⁺/VCAM1⁻ cells (Fig. 2A-C). This contrasts with robust MYOG signal detected at 5 days post injury, suggesting that YFP⁺ cells are not committed myoblasts.

To further investigate cell identity, we performed single-cell transcriptome analysis of YFP⁺ cells from middle-aged and geriatric mice and integrated these data into public single-cell sequencing data of uninjured and injured muscles (De Micheli et al., 2020, Oprescu et al., 2020) (Fig. S1A-E). Integration of public single-cell sequencing data yielded 5 clusters: quiescent MuSC, activated MuSC, myoblast/myocyte, fast fiber myonuclei and slow fiber myonuclei (Fig. 2D). The transcriptional signatures characterizing each cluster were consistent with known lineage identities, supporting the validity of the clustering (Fig. 2E). High levels of myonuclei transcripts likely represents fragmented myonuclei or cells with compromised nuclear integrity, which is a known technical byproduct of dissociation of the myofiber (Santos et al., 2021). Only high-confidence MuSC clusters were included in differential expression, trajectory inference, and cluster-proportion analyses. All YFP⁺ cells from both middle and geriatric age mapped to transcriptional clusters associated with quiescent MuSCs (Fig. 2F). This suggests that even VCAM1^{Low} and VCAM1⁻ cells from geriatric age did not cluster with activated MuSCs or differentiating myoblasts or myocytes, but rather remained within the quiescent MuSC compartment. These findings support the possibility that these cells may retain core stem-like features, despite altered surface marker profiles.

Since quiescent MuSCs are generally smaller than activated progenitors due to reduced cytoplasmic volume, low metabolic activity, and compact chromatin organization (Liu et al., 2013), we also compared the cell size of VCAM1⁻ cells with activated MuSCs from injured mice. Indeed, both VCAM1^{Low} and VCAM1⁻ cells from geriatric mice were similar in diameter to YFP⁺ cells from uninjured middle-aged mice, but were significantly smaller than activated MuSCs from injured mice (Fig. S2A-B). These results are consistent with our interpretation that VCAM1^{Low} and VCAM1⁻ cells from geriatric mice may remain in a quiescent state.

Given reports of smooth muscle mesenchymal cells (SMMCs) expressing overlapping surface markers with MuSCs, we considered whether the VCAM1⁻ cells might reflect SMMC contamination (Giordani et al., 2019). mRNA sequence analysis revealed that this population exhibited low expression of SMMC markers, including Myl9, Acta2, and Pdgfrb (Fig S2C), suggesting they are transcriptionally distinct from SMMCs. VCAM1⁻ cells showed significantly higher Pax7 expression compared to SMMCs. Together, these results suggest that YFP⁺/VCAM1⁻ cells from geriatric age may retain quiescent features and myogenic identity, despite lacking typical MuSC markers.

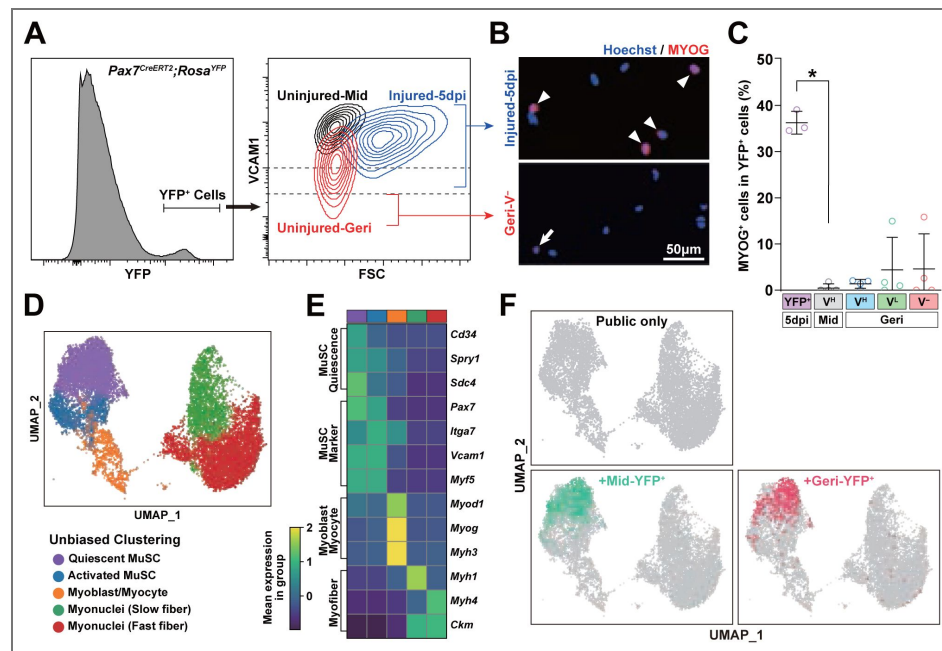


Figure 2. YFP⁺/VCAM1⁻ cells are quiescent MuSCs.

(A) Representative FACS plot for YFP⁺ cells from uninjured Mid and Geri, and injured-5dpi young (20-weeks-old) mice. (B, C) Representative images (B) of ICC staining for Hoechst (Blue) and MYOG (Red), and quantification (C) of MYOG⁺ cells in each YFP⁺ cell, one-way ANOVA followed by Bonferroni's multiple comparison test; $p = 0.0139$ between YFP⁺ cells of injured-5dpi and Mid V^H. While 5 dpi MuSCs differed significantly from young MuSCs (adjusted $p = 0.0139$), the comparisons between 5 dpi and each geriatric subgroup (VCAM-high, -mid, and -low) did not reach statistical significance after correction for multiple testing (adjusted $p = 0.17, 0.15$, and 0.17 , respectively). (D) Uniform manifold approximation and projection (UMAP) plot of YFP⁺ cells integrated with public scRNA-seq data of MuSC-lineage mononucleated cells in uninjured/injured muscle colored by cell types. (E) Matrix plot for representative genes of specific cell types for determining the identities of each cluster. (F) The localization of Mid and Geri YFP⁺ cells in this study on UMAP plot of Fig. 2D.

Impaired functionality of YFP⁺/VCAM1[−] MuSCs

Since YFP⁺/VCAM1[−] cells retained molecular and morphological features consistent with quiescent MuSCs, we next investigated their functional potential. Colony-forming assays revealed that YFP⁺/VCAM1^{High} cells from geriatric mice expanded at comparable rates to those from middle-aged mice. In contrast, colony formation by YFP⁺/VCAM1^{Low} and YFP⁺/VCAM1[−] subsets were progressively reduced, with the VCAM1[−] subsets forming smaller colonies (Fig. 3A–B). These findings suggest that these cells retain proliferative ability, although their expansion capacity was compromised.

To examine their regenerative potential *in vivo*, we transplanted equal numbers of YFP⁺ cells from each group into pre-injured TA of MuSC-deficient Pax7^{CreERT2};Rosa^{DTA} recipient mice (Fig. 3C). In the absence of donor cells, regeneration was severely impaired, consistent with a lack of endogenous MuSCs (Fig. 3D). Transplantation of YFP⁺/VCAM1^{High} cells from both middle-aged and geriatric donors supported robust regeneration, resulting in centrally nucleated myofibers with comparable morphology between age groups, consistent with heterotypic transplantation studies (Fig. 3D–E) (Ikemoto-Uezumi et al., 2015). Interestingly, both YFP⁺/VCAM1^{Low} or YFP⁺/VCAM1[−] cells were also capable of contributing to myofiber regeneration, despite lacking traditional MuSC markers (Fig. 3E). However, regenerated myofibers in these groups were significantly smaller than those derived from VCAM1^{High} cells (Fig. 3F), suggesting that loss of VCAM1 expression may be associated with diminished regenerative capacity.

Together, these results indicate that even the most marker-deficient MuSCs in geriatric muscle, YFP⁺/VCAM1[−] cells, retain a degree of regenerative competence. However, their reduced proliferative output and attenuated *in vivo* contribution suggests functional heterogeneity within the aged MuSC pool, wherein diminished marker expression may correlate with impaired regenerative performance.

VCAM1[−] MuSCs exhibit senescence-like features and sensitivity to senolytics

To investigate the molecular basis of impaired function in VCAM1[−] MuSCs, we performed mRNA sequence analysis on YFP⁺ MuSCs. Samples showed high quality and a strong correlation was observed among biological replicates (Fig. S3A–C). Each group consistently reproduced similar gene expression patterns in biological replicates in hierarchical clustering (Fig. S3D). Although mid-aged VCAM1-high MuSCs did not exhibit the emergence of a VCAM1-low state, mRNA profiling revealed detectable transcriptional differences between mid-aged and geriatric VCAM1-high populations (Fig. S3D). These shifts likely reflect the gradual accumulation of aging-associated molecular changes that begin during the middle-aged stage, even before overt functional decline or loss of VCAM1 becomes evident. Importantly, these early transcriptomic alterations do not give rise to the dysfunctional VCAM1-low subset, which emerges only in geriatric muscle. In particular, hierarchical clustering showed similar gene expression pattern between VCAM1^{High} MuSCs of middle-aged and geriatric mice and between VCAM1^{Low} and VCAM1[−] MuSCs (Fig. S3E), consistent with our results of *in vivo* and *in vitro* functionality experiments (Fig. 3A–F).

Kyoto-encyclopedia of genes and genomes (KEGG) pathway enrichment analysis based on the DEGs (Fig. S3E) (Kolberg et al., 2023) showed alterations in pathways related to cellular senescence and PI3K-AKT pathway in VCAM1[−] MuSCs of geriatric age as compared to VCAM1^{High} MuSCs of middle-age (Fig. 4A). We further investigated the expression levels of individual genes known to be altered in MuSCs of aged mice, such as genes related to extracellular matrix (Schuler et al., 2021), Notch signaling (Liu et al., 2018; Conboy et al., 2003; Conboy et al., 2005), and autophagy-lysosomal pathway (Garcia-Prat et al., 2016; Kim et al., 2021b), some of which were more altered in VCAM1^{Low} and VCAM1[−] MuSCs compared to VCAM1^{High} MuSCs of geriatric mice (Fig. 4B). *Cdkn2a* (p16^{INK4a}) and *Cdkn2b* (p15^{INK4b}), which are cell cycle regulators widely used as markers for senescent cells, showed highly altered expression patterns in VCAM1[−] MuSCs compared to VCAM1^{Low} MuSCs (Fig. 4B) (Saito and Chikenji, 2021). However, the expression

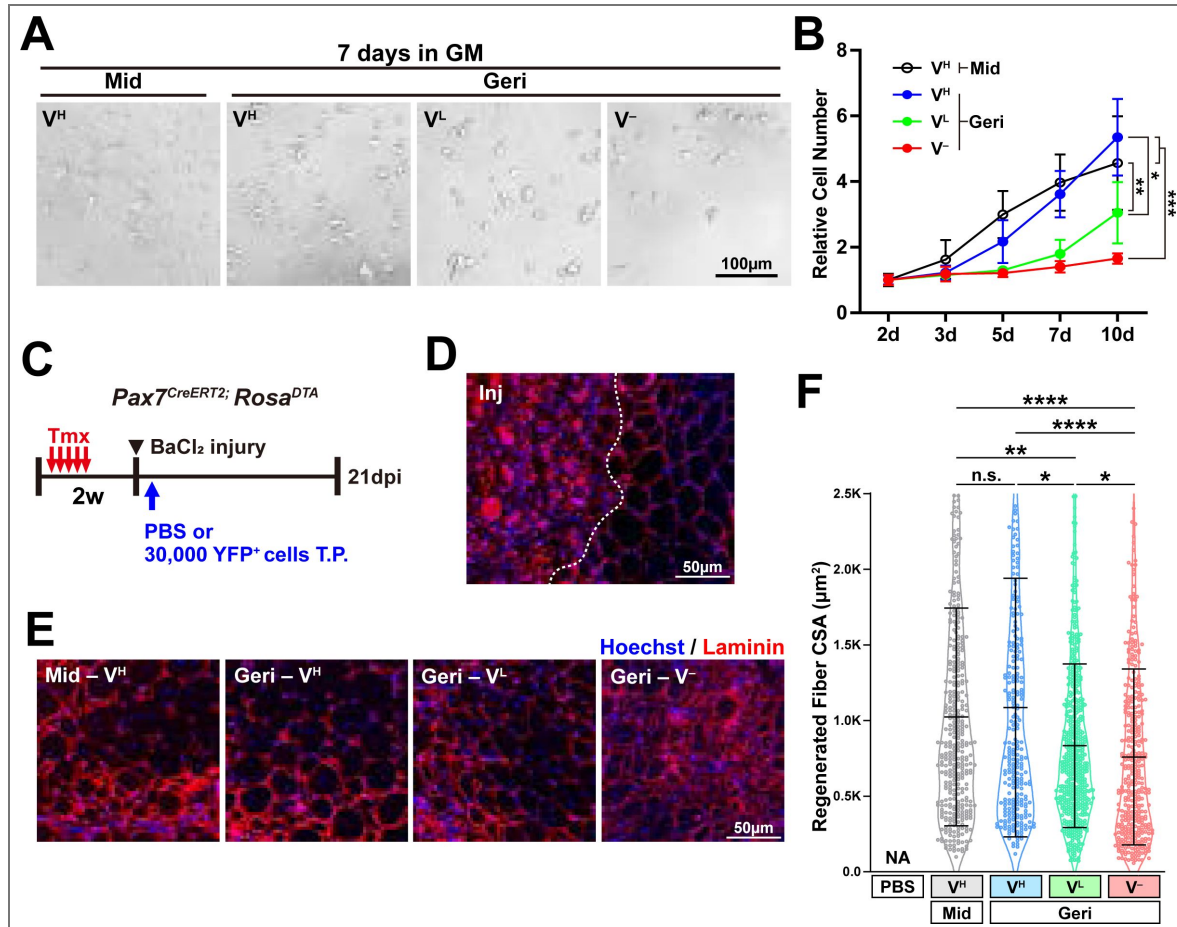


Figure 3. Compromised but retained functionality in VCAM1⁻ MuSCs.

(A) Representative bright field images of Mid V^H , Geri V^H , V^L , and V^- MuSCs at 7 days culture in MuSC growth media. (B) Proliferation measured by relative cell number at 2, 3, 5, 7, and 10 days, two-way ANOVA followed by Bonferroni's multiple comparison test; $p = 0.0029$ and 0.0002 (Mid V^H vs V^L and V^- , respectively) and $p = 0.0102$ and 0.0006 (Geri V^H vs V^L and V^- , respectively). For each sample, 30,000 MuSCs were plated at day 0, with 3 biological replicates per condition. (C) Experimental schematics of transplantation of YFP⁺ cells to TA muscle of $Pax7^{CreERT2}; Rosa^{DTA}$ mice. (D, E) Representative images of IHC staining for Hoechst (Blue) and Laminin (Red) in PBS treated TA muscle section following injury. (D) Left side of the dotted line show areas affected by BaCl₂ injury and right side shows uninjured areas of the muscle. (E) IHC staining shows centralized myofibers in all images, with notably smaller myofibers in Geri V^- . (F) Quantification of cross-sectional area (CSA) of regenerated fibers from TA muscle of $Pax7^{CreERT2}; Rosa^{DTA}$ mice transplanted with Mid V^H , Geri V^H , V^L , and V^- at 21 dpi, $n = 3$ for each group. For each mouse, the entire TA muscle was cryosectioned and immunostained, and all regenerated fibers containing centrally located nuclei were included in the CSA quantification. Total 43 data points are outside of the axis. Kruskal-Wallis test followed by Dunn's multiple comparison test; $p < 0.0001$ (Mid and Geri V^H vs V^-) and $p = 0.0059$ (Mid V^H vs Geri V^L), 0.0149 (Geri V^H vs V^L) and 0.0143 (Geri V^L vs V^-).

levels of other well-known cellular senescence marker genes, such as *Rb1*, *Il6*, and *Trp53*, were not altered much. *Pik3cb* and *Tsc1*, which are involved in the PI3K-Akt-mTOR pathway, showed overall increased expression, while other genes in the PI3K-Akt pathway remained relatively unchanged. Together, these results suggest a unique combination of *in vivo* senescence-like molecular characteristics of VCAM1[−] MuSCs, distinct from molecular characteristics of replicative cellular senescence.

Given their unique senescence associated transcriptome and partially elevated PI3K-Akt pathway, we tested whether VCAM1[−] MuSCs were susceptible to senolytic elimination. Geriatric mice were treated with senolytic agents dasatinib and quercetin (DQ) (Xu et al., 2018 [4](#)), and analyzed 2 weeks later (Fig. 4C [4](#)). Despite variation in baseline VCAM1^{High} and VCAM1^{Low/−} population across aged animals, a consistent inverse correlation was observed between VCAM1^{High} and VCAM1^{Low/−} fractions (Fig. S3F). The multimodal distribution observed in the geriatric group reflects increased mouse-to-mouse variability that emerges at very advanced ages. Following DQ treatment, VCAM1^{High} MuSCs remained comparable while VCAM1^{Low/−} MuSCs showed reduced abundance and lower variability ($p = 0.0596$) (Fig. 4D [4](#)). The multimodal distribution observed in the geriatric group reflects increased mouse-to-mouse variability that emerges at very advanced ages. Although aging affects all animals, the extent and timing of phenotypic changes vary substantially among individuals, resulting in broader divergence in VCAM1 expression states. Statistical analyses were performed using the mouse as the biological replicate, this variability does not alter the overall conclusion. These results indicate selective sensitivity of VCAM1^{Low/−} MuSCs to senolytic agents and their increased dependence on BCL-2-related survival pathways. Together, our findings reveal a distinct subset of quiescent MuSCs in geriatric muscle that exhibit unique senescence-like molecular features and are amenable to senolytic clearance. We term these cells GERI-MuSCs (Geriatric Emerging Regeneration-capable and Impaired-MuSCs).

Rejuvenation of GERI-MuSCs by DHT treatment

Studies have demonstrated that treatment with rejuvenating agents can restore myofiber regeneration capacity in aged mice (Bernet et al., 2014 [4](#), Cosgrove et al., 2014 [4](#), Zhang et al., 2016 [4](#), Elabd et al., 2014 [4](#), Price et al., 2014 [4](#), Garcia-Prat et al., 2016 [4](#)). Previously, dihydrotestosterone (DHT) supplementation has been shown to preserve MuSCs function and enhance regeneration in Antide, a GnRH antagonist, treated mice by restoring autophagy activation (Kim et al., 2021b [4](#)). Consistent with these findings, implantation of DHT-filled silastic tube in 28-month-old geriatric mice improved muscle regeneration, as evidenced by increased fiber cross sectional area (Fig. S4A-C).

To investigate whether DHT could affect GERI-MuSCs, we treated 28-month-old MuSC^{YFP} mice with DHT and analyzed MuSC populations one month later (Fig. 4E [4](#)). Remarkably, DHT treatment resulted in a decrease in GERI-MuSCs with a corresponding expansion of VCAM1^{High} MuSCs population. This phenotypic reversion suggests that DHT promotes restoration of canonical marker expression in GERI-MuSCs.

To determine whether DHT acts through rejuvenation of GERI-MuSCs, we first treated the mice with DQ for 5 consecutive days and then implanted DHT tube 2 weeks later (Fig. 4F [4](#)). In mice pre-treated with DQ, DHT supplementation failed to reconstitute the VCAM1^{High} MuSC pool (Fig. 4F [4](#)). This finding suggests a model in which GERI-MuSCs are not terminally senescent, but rather exist in a latent, hormone-responsive state. Upon DHT exposure, these cells likely reacquire a VCAM1^{High} phenotype, underscoring their potential as a therapeutic target in muscle aging.

Identification of GERI-MuSCs using CD63 and CD200

In non-reporter models, the detection and isolation of GERI-MuSCs pose a significant challenge, as these cells lack expression of conventional MuSC markers such as VCAM1 and PAX7 at geriatric age. To overcome this limitation, we conducted a transcriptomic screen to identify alternative surface markers that remain stably expressed across the full spectrum of MuSC aging states. Candidate genes were selected based on three criteria; high transcript abundance (RPKM),

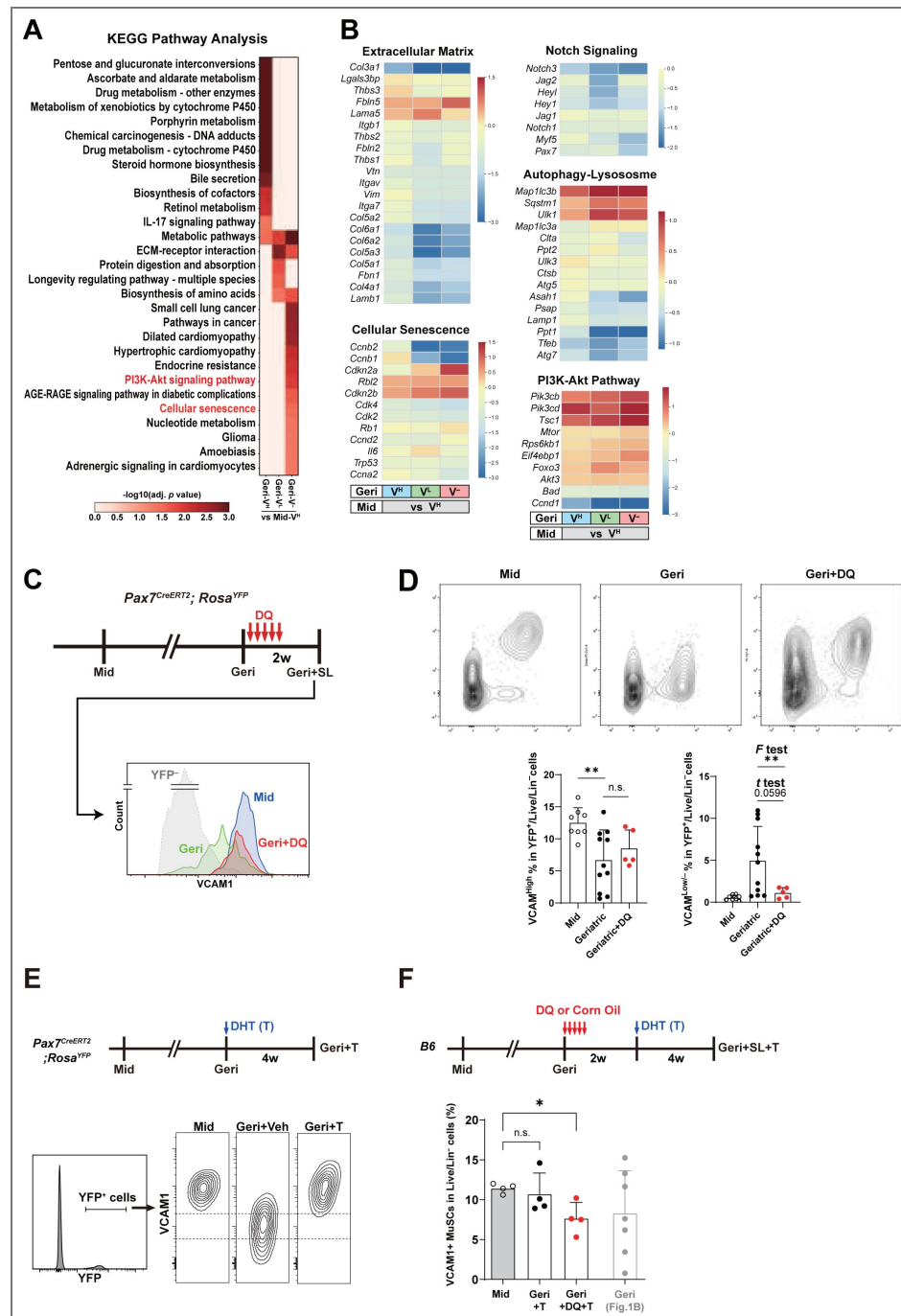


Figure 4. Senescence characteristics of VCAM1⁻/YFP⁺ cells and their rejuvenation by DHT supplementation.

(A) The statistical significance of Kyoto-Encyclopedia of Genes and Genomes (KEGG) pathways, associated with differentially expressed genes (DEGs) between Mid V^H and Geri V^H , V^L and V^- , visualized as a heatmap. (B) Heatmap of relative expressions of representative genes known to be altered in aged MuSCs compared to Mid V^H . (C) Experimental schematics of senolytics, dasatinib and quercetin (DQ), treatment in MuSC^{YFP} at geriatric age and representative FACS histogram of VCAM1 expression. Mid, Geri, and Geri treated with SL (Geri+DQ) mice. (D) Representative FACS plots and quantification of the percentage of VCAM1^{High} and VCAM1^{Low} cells divided by VCAM1 expression levels in Mid (n = 8) and Geri mice with (n = 5) or without (n = 11) senolytics, 2 tailed student's t test, $p = 0.0051$ for Mid vs Geri in VCAM1^{High} and F test: $p = 0.0026$ for Geriatric vs Geriatric+DQ in VCAM1^{Low}. (E) Experimental schematics of dihydrotestosterone (DHT) treatment and FACS plot of VCAM1 expression levels for YFP⁺ cells in MuSC^{YFP}. (F) Experimental schematics of DQ and DHT treatment and quantification of the percentage of VCAM1^{High} MuSCs in Live/Lin⁻ cells in Mid and Geri B6 mice (n = 4 in each), 2 tailed student's t test, $p = 0.0113$ for Young vs Geri+DQ+T.

consistent expression from VCAM1^{High} to VCAM1⁻ MuSCs, and predicted surface localization (Fig. 5A). CD63 and CD200 sufficed all three criteria and were therefore chosen as candidate markers for further validation (Fig. 5B).

We confirmed protein level expression of CD63 and CD200 expression in MuSCs by co-immunostaining TA cross-sections for CD63, CD200, YFP and PAX7. Both markers co-localized with YFP and PAX7 beneath the basal lamina (Fig. 5C). Flow cytometry analysis further demonstrated that 96.3% of YFP⁺/VCAM1⁺ cells from young MuSC^{YFP} mice were CD63^{high}/CD200^{high} (Fig. 5D). Notably, among lineage-negative (CD31⁻/CD45⁻), SCA-1⁻, and live cells, 93.2% of CD63^{high}/CD200^{high} cells were YFP⁺/VCAM1⁺ cells (Fig. 5E). Importantly, 96.5±2.8% of sorted CD63^{high}/CD200^{high} cells were PAX7⁺, validating these markers as a robust strategy for MuSC enrichment.

To assess whether the CD63/CD200-based strategy could detect and isolate GERI-MuSCs in wild-type mice, we applied the strategy to middle and geriatric aged mice. In middle aged mice, all CD63^{high}/CD200^{high} cells corresponded exclusively to VCAM1^{High} MuSCs (Fig. 5F). In contrast, geriatric mice exhibited a marked shift, with the majority of CD63^{high}/CD200^{high} cells falling within the VCAM1^{Low/-} gate (Fig. 5F). These results indicate that CD63 and CD200 expression is preserved in GERI-MuSCs, enabling their prospective identification and isolation in the absence of genetic labeling. Therefore, these findings establish CD63 and CD200 as reliable surface markers that can be used with VCAM1. Together, they can detect the full spectrum of MuSCs, including functionally latent populations that evade detection by conventional means. This strategy offers a critical tool for future studies aiming to comprehensively characterize MuSC heterogeneity and therapeutic responsiveness in the context of aging.

Discussion

Our study challenges the longstanding paradigm that muscle stem cell (MuSC) number declines irreversibly with age. While previous reports have described significant depletion of Pax7⁺ MuSCs in aged muscle, these findings were largely based on immunohistochemical detection or flow cytometric profiling using conventional surface markers. Here, we demonstrate that genetically labeled MuSCs, tracked longitudinally *in vivo* using a Pax7-CreERT2;Rosa-YFP model, do not decrease in number even in geriatric mice. Instead, a substantial subset of aged MuSCs loses expression of canonical markers such as Pax7 and VCAM1, rendering them undetectable by traditional approaches. Although VCAM1 is frequently used as a surface marker for MuSC identification, it represents only one of several commonly used markers, and ITGA7-based strategies remain widely applied in the field. In our dataset, ITGA7 expression within the YFP⁺ MuSC population did not show a significant reduction in mean fluorescence with aging (Fig. 1D). However, the variance of ITGA7 expression was markedly increased in geriatric MuSCs, indicating greater phenotypic heterogeneity. This pattern suggests that aging does not uniformly downregulate canonical markers but instead increases instability and variability in their expression, which may lead to under-detection of a subset of MuSCs even when ITGA7 is used as the primary gating marker. These findings underscore that marker-based quantification may underestimate specific aging-associated MuSC subpopulations and should be interpreted within this limitation.

We identify this overlooked population as GERI-MuSCs (Geriatric Emerging Regeneration-capable and Impaired MuSCs). Despite their diminished expression of hallmark MuSC markers, GERI-MuSCs exhibit features consistent with quiescence, rather than terminal differentiation or apoptosis. Although GERI-MuSCs showed markedly reduced colony formation and further validation using direct viability assays are necessary, several lines of evidence suggest that they are viable rather than acutely undergoing cell death. These cells can be isolated as intact populations by FACS and generate high-quality RNA-sequencing libraries, which would not be feasible if they were non-viable. Their transcriptomic profiles also show enrichment of stress-response and senescence-associated programs rather than apoptotic signatures. Together, these findings suggest that GERI-MuSCs remain alive but exhibit impaired proliferative capacity. Furthermore, transcriptomic profiling and functional assays reveal that these cells retain a partial

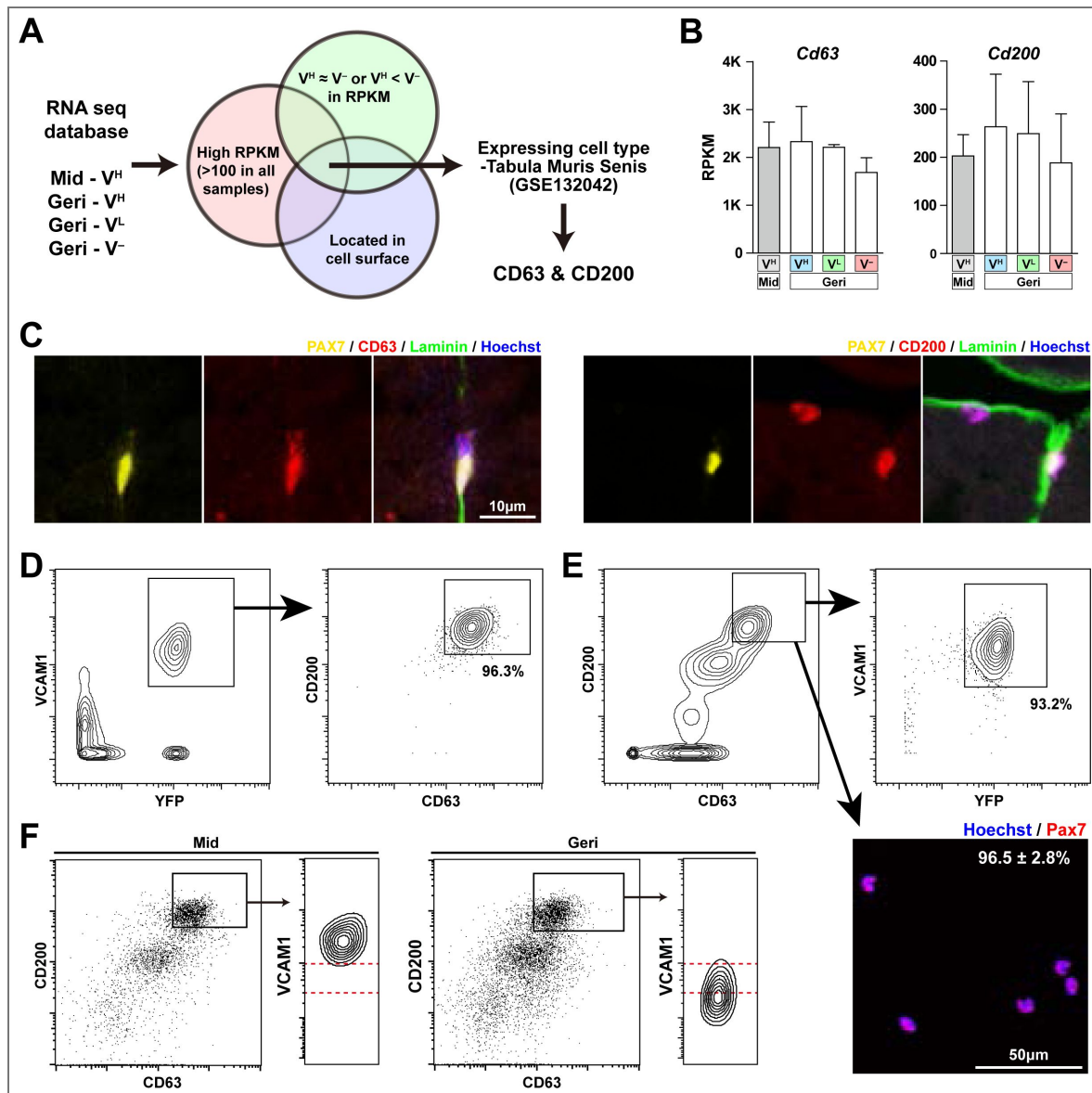


Figure 5. Detection of GERI-MuSCs using CD63/CD200-based FACS strategy.

(A) Venn Diagram schematics showing criteria used to select MuSC marker candidate CD63 and CD200, from RNA seq database. (B) Relative gene expressions of *Cd63* and *Cd200* shown in RPKMs from Mid V^H , Geri V^H , V^L , and V^- MuSCs. (C, D) Representative IHC staining images for Hoechst (Blue), Laminin (Green), PAX7 (Yellow) with CD63 (red, C) or CD200 (red, D) in the TA muscle of Mid MuSC^{YFP}. (E) Representative FACS plot from young mice showing 96.3% of VCAM1⁺/YFP⁺ cells are CD63^{high}/CD200^{high}. (F) Left: representative FACS plot from young mice showing 93.2% of CD63^{high}/CD200^{high} cells are VCAM1⁺/YFP⁺, and ICC staining of this population shows that 96.5 $\pm$ 2.8% are PAX7⁺. Right: representative FACS plot of MuSCs of B6 Mid and Geri mice gated by CD63 and CD200 and their VCAM1 expression level.

capacity for proliferation and regeneration, albeit at reduced efficiency. When comparing geriatric VCAM1⁺ MuSCs with middle aged MuSCs, we found 1,428 DEGs, where 701 genes were downregulated and 727 genes were upregulated (Fig. S3E). Some of the pathways altered were similar to previously reported differences, such as alterations in the autophagy-lysosome related genes and PI3K-Akt Pathways (Garcia-Prat et al., 2016 [↗](#)). However, these transcriptomic alterations did not affect the functional integrity of geriatric VCAM1⁺ MuSCs (Fig. 3 [↗](#) A-F). Taken together, these results do not contradict prior studies demonstrating age-related MuSC decline but rather refine this understanding by highlighting a subset of aged MuSCs that may have been underrepresented or more difficult to detect using conventional markers. By resolving these cells with an alternative phenotypic definition, our findings complement existing models of MuSC aging and suggest that part of the apparent depletion previously reported may reflect changes in marker expression rather than complete cellular loss.

GERI-MuSCs display a unique gene expression profile that includes upregulation of senescence-associated transcripts such as *Cdkn2a* but unaltered expression of other typical senescence-associated genes, such as *Il6* and *Trp53*, while showing preservation of certain stemness-related pathways (Fig. 4B [↗](#)). These cells do not fulfill the classical definition of cellular senescence, such as irreversible cell cycle arrest (Campisi and d'Adda di Fagagna, 2007 [↗](#)). p16 expression is elevated in VCAM-low MuSCs, but this alone does not establish a fully senescent state. Rather, GERI-MuSCs exhibit a hybrid state of retaining regenerative potential while expressing select markers traditionally associated with senescence. This finding suggests that the criteria used to define senescence in proliferative cell types may not fully apply to quiescent stem cells like MuSCs, which spend most of their life in a non-proliferative state. Additional markers and functional assays will be needed to define the precise state of these cells and to delineate *in vivo* senescence features specific to long-lived, tissue-resident stem cells.

Our results suggest that GERI-MuSCs could be functionally reactivated. Treatment with dihydrotestosterone (DHT) restored VCAM1 expression in GERI-MuSCs and regenerative performance in aged mice, suggesting that their phenotype is reversible and hormonally responsive. This provides a compelling explanation for the efficacy of hormone-based interventions in aging muscle and highlights the potential for targeting latent regenerative capacity in aged stem cells. Notch signaling appears to be a key regulator of this reversible phenotype. We found diminished expression of Notch pathway components, including Notch1, Notch3, Hey1, and HeyL, in VCAM1^{Low} and VCAM1⁻ subsets, along with reduced transcription of Pax7 and VCAM1. Given that both Pax7 and VCAM1 are downstream targets of Notch signaling and that Notch activity can be modulated by androgen signaling, the restoration of MuSC function by DHT may act through this axis. Testing such mechanisms in future studies will be important for establishing therapeutic strategies.

Although DHT supplementation partially restored the regenerative performance of geriatric MuSCs, this effect cannot be attributed solely to the rescue of VCAM1 expression in GERI-MuSCs. Androgen signaling is known to influence multiple non-stem cell populations in the muscle environment, including fibro-adipogenic progenitors, myofibers, and immune cells, any of which may contribute to the observed improvement in regeneration (Kim et al., 2021b [↗](#)). Accordingly, we acknowledge that DHT likely exerts broader tissue-level effects beyond the MuSC compartment, and our findings should be interpreted within this limitation. Future studies combining senolytic strategies with targeted modulation of androgen signaling may help delineate the relative contributions of each pathway, although such experiments extend beyond the scope of the present work.

The presence of GERI-MuSCs also raises caution about the use of senolytic agents. Although dasatinib and quercetin are used to clear senescent cells and have shown benefits in aged tissue, our data show that these agents also deplete GERI-MuSCs. Since GERI-MuSCs retain functionality and regenerative potential, their removal could compromise muscle maintenance and repair in elderly individuals. Given the critical role of MuSCs in preserving muscle mass, supporting myonuclei turnover, and maintaining neuromuscular junctions, the long-term application of senolytics in skeletal muscle warrants careful scrutiny.

Finally, we propose CD63 and CD200 as reliable surface markers that can be used with VCAM1 for identifying MuSCs across age groups. These markers remain expressed in both young and aged MuSCs, including those with diminished canonical marker expression. CD63 and CD200 expression in geriatric muscle was evaluated using conventionally isolated MuSCs rather than YFP-traced cells. This distinction should be considered when interpreting these results. Their use enables comprehensive isolation of the full MuSC population and will be instrumental in improving the resolution of future studies on stem cell heterogeneity and aging.

In conclusion, we identify a previously unrecognized population of MuSCs—GERI-MuSCs—that persists with age despite loss of conventional markers. These cells remain quiescent, partially functional, and responsive to rejuvenation cues. Their discovery challenges existing models of MuSC depletion and senescence, and underscores the importance of revisiting stem cell aging with improved detection strategies and nuanced interpretations of cellular function.

Materials and Methods

List of Antibodies for Flow Cytometry and FACS (Fluorescence-activated cell sorting)

Marker	Fluorophore	Catalog No.	Company
CD31	APC	102510	BioLegend
CD31	Pacific Blue™	108120	BioLegend
CD45	APC	103112	BioLegend
CD45	Pacific Blue™	405206	BioLegend
VCAM1	Biotin	553331	BD Biosciences
Streptavidin	PE/Cy7	405206	BioLegend
Ly-6A/E (Sca-1)	Pacific Blue™	108120	BioLegend
Ly-6A/E (Sca-1)	FITC	553335	BD Biosciences
CD200	BV711	745548	BD Biosciences
CD63	APC	170631-82	eBioscience

Animals

All animal procedures conformed to the guidelines of Institutional Animal Care and Use Committee (IACUC) of Seoul National University, Korea. Male C57BL/6J mice were housed under a 12-h light/12-h dark cycle at 22°C with 40-60% humidity, and fed ad libitum. Pax7^{CreER} (017763), Rosa^{YFP} (006148), Rosa^{DTA} (009669) mice were obtained from the Jackson Laboratory (Bar, Harbor, ME, USA). Tamoxifen (25mg/mL in corn oil; Sigma-Aldrich, St. Louis, MO, USA) was administered orally for 5 consecutive days (240 mg/kg). For muscle injury, mice were anesthetized with avertin (250 µg/g, i.p.) and injected with 60 µL of 1.2% BaCl₂ (Sigma-Aldrich) into the Tibialis Anterior (TA) muscles. One-day post-injury, mice were anesthetized with avertin and MuSCs were transplanted at indicated numbers.

Flow Cytometry and FACS (Fluorescence-activated cell sorting)

Limb muscles were mechanically dissected and enzymatically dissociated in Dulbecco's modified Eagle's medium (DMEM) containing 10% horse serum (Hyclone, Logan, UT, USA), collagenase type II (800U/mL; Worthington, Lakewood, NJ, USA), and dispase (1.1 U/mL; StemCell Technologies, Vancouver, BC, Canada) at 37°C for 50 minutes. Digested suspensions were triturated, washed with DMEM and filtered to obtain mononuclear cells. Cells were stained with antibodies, analyzed and sorted using a FACS Aria III or FACS Aria Fusion (BD Bioscience). List of antibodies used are provided as supplementary materials. For all flow cytometry analyses, fibro-adipogenic progenitors (FAPs) were excluded by gating out Sca1⁺CD31⁻CD45⁻ cells prior to MuSC identification. This strategy ensured that YFP⁺ MuSC quantification was not confounded by age-associated changes in Sca1⁺ mesenchymal populations. Representative plots were processed with FlowJo v10.9.

MuSC Culture

FACS-isolated MuSCs were cultured in DMEM/F12 + 20% FBS + 1% ZellShield (Minerva Biolabs, Germany) + 2.5 ng/mL bFGF (F0291, Sigma Aldrich) on Matrigel (256231, BD Bioscience)-coated plates at 37°C and 5% CO₂. Medium was refreshed every 3 days. MuSCs were imaged daily using EVOS FL Auto 2 (Thermo Fisher, USA) and the number of cells in the same region was counted.

Histochemistry and Immunohistochemistry

TA muscles were fixed overnight in 4% paraformaldehyde, embedded in the Tissue Tek OCT compound (Sakura Finetek, USA), and snap-frozen in liquid nitrogen. The muscles were sectioned at 7µm and stained on the slides. H&E staining was performed according to basic protocols. For immunostaining, sections were subjected to epitope retrieval in 50mM Tris-EDTA (pH 9.0) at 105°C for 10 minutes, then blocked with 3% BSA + 0.1% Triton X-100. Sections were incubated overnight with primary antibodies, mouse anti-PAX7 (DSHB, Pax7, 1:100 dilution), chicken anti-GFP (abcam, ab13970, 1:300), rabbit anti-laminin (Sigma, L9393-2ML, 1:500), rat anti-CD200 (BD Biosciences, 1:50), and rat anti-CD63 (eBioscience, 1:50). After washing, sections were incubated with secondary antibodies, Alexa Fluor 647-conjugated anti-rat or anti rabbit IgG, Alexa Fluor 594-conjugated anti mouse IgG, and Alexa Fluor 488-conjugated anti chicken IgG. Nuclei were counterstained with Hoechst 33342 (Invitrogen, H3570). Slides were mounted with VECTASHIELD (Vector Laboratories) and images were acquired using EVOS FL Auto 2 (Thermo Fisher) and Leica TCS SP8 confocal microscope. Cross-sectional area (CSA) was calculated by SMASH[23] and ImageJ programs, respectively.

Immunocytochemistry

Isolated MuSCs were cytospun onto slides (Cytospin 4, Thermo Scientific), fixed with 4% paraformaldehyde for 10 minutes, washed with PBS and permeabilized with PBS containing 0.4% Triton X-100 for 30 minutes. Slides were then incubated with primary antibodies, mouse anti-PAX7 (DSHB) and mouse anti-MYOG (DSHB, F5D, 1:100 dilution) in blocking buffer. Subsequent steps followed the immunohistochemistry protocol. Relative fluorescent intensity was quantified using ImageJ.

Senolytics and DHT treatment

Dihydrotestosterone (DHT) implantation was performed as previously described[8]. Senolytic agents Dasatinib (D-3307, LC laboratories, 5mg/kg) and Quercetin (Q4951, Sigma, 50 mg/kg) was administered orally for 5 consecutive days. Experimental schemes of combined treatment are as mentioned in the results section of this article.

Single-cell RNA sequence analysis

To obtain gene-specific UMI (Unique Molecular Identifier) counts in each cell, we processed raw scRNA-seq data using the BD Rhapsody WTA analysis pipeline (v1.11.1)[24]. Scanpy (v1.9.3)[25] was utilized for single-cell RNA sequence analysis. Cells with a gene count below 200 or a mitochondrial gene percentage above 20% were considered low-quality cells and were excluded from downstream analyses. We set the upper limit of each UMI count as $(\text{mean}(\text{total counts}) + 2 \text{ standard deviation}(\text{total counts}) \text{ value})$ to remove doublets. Normalization was applied to the count matrix (`scanpy.pp.normalize_total` with `target_sum=104` and `scanpy.pp.log1p`).

We characterized the enriched cells by merging our scRNA-seq data with public datasets (GSE162172, GSE138826)[26,27]. From the public dataset, only the cells belonging to the myogenic lineage (from quiescent MuSCs to Myonuclei (Type IIb or IIx)) were selected for comparison. After merging the public dataset with our dataset (`scanpy.external.pp.harmony_integrate` with `max_iter_harmony=50`), 2000 highly variable genes were identified and scaled (`scanpy.pp.highly_variable_genes` with `n_top_genes=2000`, `scanpy.pp.scale` with `max_value=10`).

We visualized the scRNA-seq data using uniform manifold approximation and projection (UMAP) (`scanpy.tl.umap`) with a precomputed neighborhood graph (`scanpy.pp.neighbors` with `n_pcs=50`). We clustered the UMAP-transformed data using the Louvain algorithm (`scanpy.tl.louvain`, `resolution=0.5`). Each cluster was annotated using cell-specific markers (Fig. 1D, E [↗](#)). The proportions of quiescent MuSCs, activated MuSCs, and Myoblasts/Progenitors were measured in our samples (Mid and Geri) and the public dataset across various days-post-injury (DPI). Cases with less than 50 cells per DPI were excluded during this analysis (Fig. S1A).

We utilized diffusion map, PAGA, and force-directed graph to analyze the trajectory of cells belonging to the myogenic lineage. After dimension reduction using diffusion maps (`scanpy.tl.diffmap`), we computed neighborhood graphs (`scanpy.pp.neighbors`) and abstracted the graphs using PAGA (`scanpy.tl.paga`). We modeled the trajectory of the cells using the abstracted graphs as initial positions by constructing a force-directed graph (Fig. S1B-E).

Representative Isoforms of Mouse mRNAs

To acquire a sole representative isoform for each gene, we utilized the filtering pipeline described in a previous study (Kim et al., 2016 [↗](#)). The mouse reference annotation (GRCm39 refFlat) was downloaded from the UCSC website (Pruitt et al., 2007 [↗](#)). mRNA sequences with accession category 'NM' located on the 21 chromosomes (1-19, X, and Y) were selected for downstream analyses. Entries with incorrect start or stop codons, internal stop codons, and ORF lengths not divisible by 3, were removed. Nonsense-mediated mRNA decay candidates, defined as the mRNAs with stop codons positioned less than 50 nucleotides before the final exon-exon junction, were excluded (Nagy and Maquat, 1998 [↗](#)). A total of 20,582 non-redundant representative isoforms were selected as mouse reference mRNAs for subsequent analyses.

mRNA sequence analysis

mRNA sequence analysis was performed on 3-4 biological replicates for each condition. We trimmed low-quality read ends using a custom algorithm. Each base position was assigned a reward (+5) or a penalty (-1) based on its Phred score with a threshold of 30. To define high-quality read ends, we calculated the cumulative sum from both ends, identified the positions with the minimum value, and trimmed the reads at those positions. This process was repeated iteratively until no further trimming was necessary. Reads shorter than 17 bases after trimming were excluded. Subsequently, we removed adapter sequences using Cutadapt (v3.1) (Martin, 2011 [↗](#)) with options (-O 10 -m 18 --match-read-wildcards). We eliminated sequencing artifacts and PCR duplicates using FASTX-Toolkit's `fastx_artifact_filter` with option (-Q 33) (http://hannonlab.cshl.edu/fastx_toolkit/ [↗](#), v0.0.13). To eliminate reads derived from mouse non-coding RNA, we aligned the reads to non-coding RNA sequences in Rnacentral (Consortium,

2021) using Bowtie2 (v2.1.0) (Langmead and Salzberg, 2012) with options (-k 1 --norc --very-sensitive), and the reads aligned to non-coding RNAs were excluded. Unpaired reads were also excluded.

Preprocessed reads were aligned to the mouse genome (GRCm39) utilizing the STAR alignment software (v2.7.5b) (Dobin et al., 2013). The parameters used for the STAR alignment are described in a previous study (Kim et al., 2021a). The reads were sorted and indexed using Samtools (v0.1.19) (Danecek et al., 2021). The expression of each gene was quantified by calculating the fragment per kilobase of transcript per million mapped reads (FPKM).

We applied the same analysis pipeline to both our experimental samples (Mid, V^H , V^L , V^-) and publicly available SMMC samples (GSE110878, SRR67542132-SRR67542134) (Giordani et al., 2019) to calculate gene expression levels. Quantile-normalized expression levels were compared between the two datasets (Fig. S2C).

The following analyses were performed exclusively on our experimental samples.

To assess the quality of our samples, the proportion of reads that were either removed, unmapped, or successfully mapped was computed for each sample throughout the preprocessing and alignment steps (Fig. S3B). The correlation between replicates was quantified using Spearman's R (Fig. S3C).

We normalized all samples using the following procedure to ensure a fair inter-sample comparison. We applied quantile normalization to equalize the FPKM distribution of all samples. To remove individual mouse variations, we used the RUVs function ($k=2$) from the RUVSeq package (v1.34.0) (Risso et al., 2014). Subsequently, genes with normalized expression higher than the 50th percentile were selected, and the expression values were z-normalized. We clustered the samples using the clustermap function of the Seaborn package (v0.11.1) (Waskom, 2021) with correlation metric (Fig. S3D).

We identified differentially expressed genes (DEGs) by comparing the expression between the Mid, V^H , V^L , and V^- samples. We used DESeq2 (v1.40.2) (Love et al., 2014) to compute $\log^2$ fold changes and false discovery rate (FDR) corrected Q values for individual genes. DEGs were defined as genes with $|\log^2 \text{ fold change}| > 1.2$ and FDR-corrected Q value < 0.05 (Fig. S3E).

To annotate the pathways linked to DEGs, we performed enrichment analysis for Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways using the g:Profiler web server (Kolberg et al., 2023; Kanehisa et al., 2016). Among the highly expressed 12,000 genes, DEGs were queried using the g:GOST tool, and the g:SCS algorithm was employed for multiple testing correction. Pathways with multiple testing corrected Q value < 0.05 were presented (Fig. 3A).

Statistical Analysis

All statistical analyses were performed using R (R-4.3.1) and GraphPad Prism. All variables were confirmed to have a normal distribution by using the Shapiro-Wilk test and Kolmogorov-Smirnov test. For comparison of significant differences in 2 groups of normally distributed data, Levene's test for homogeneity of variance using the car package followed by a 2-tailed Student's t test or Welch's t test was used. For comparison of multiple groups, one-way analysis of variance (ANOVA) or two-way ANOVA followed by Bonferroni's multiple comparison test was performed. All data are presented as mean $\pm$ s.d. n.s., not significant; $P > 0.05$; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$. P value of < 0.05 considered statistically significant at the 95% confidence level. The number of biological replicates, p values, and detailed statistical method for each data were indicated in the figure legends.

Data availability

The bulk RNA sequencing data and scRNA-seq data have been deposited at Gene Expression Omnibus (GEO, GSE accession number: GSE254475 and GSE254476, respectively) and are publicly available as of the date of publication. Source data supporting the findings of this study are

provided with this paper. Additional data that support the findings of this study are available from the corresponding author upon reasonable request.

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Additional files

[Supplementary Figures and Legends](#) 

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Peer reviews

Reviewer #1 (Public review):

It is widely accepted that the number of muscle stem cells (MuSCs) declines with aging, leading to diminished regenerative capacity. In this study, when MuSCs were labeled with YFP at a young age, the authors found that the YFP-positive MuSC population remained stable with aging. However, VCAM1 and Pax7 expression levels were reduced in the YFP-positive MuSCs. These VCAM1-negative/low cells exhibited limited proliferative potential and reduced regenerative ability upon transplantation into MuSC-depleted mice. Furthermore, Vcam1-/low MuSCs were highly sensitive to senolysis and represented the population in which Vcam1 expression could be restored by DHT. Finally, the authors identified CD200 and CD63 as markers capable of detecting the entire geriatric MuSC population, including Vcam1-/low cells. Although numerous studies have reported an age-related decline in MuSC numbers, this study challenges that consensus. Therefore, the conclusions require further careful validation.

Major comments:

(1) As mentioned above, numerous studies have reported that the number of MuSCs declines with aging. The authors' claim is valid, as Pax7 and Vcam1 were widely used for these observations. However, age-related differences have also been reported even when using these markers (Porpiglia *et al.*, *Cell Stem Cell* 2022; Liu *et al.*, *Cell Rep* 2013). When comparing geriatric Vcam1⁺ MuSCs with young MuSCs in this study, did the authors observe any of the previously reported differences? Furthermore, would increasing the sample size in Figure 1 reveal a statistically significant difference? The lack of significance appears to result from variation within the young group. In addition, this reviewer requests the presentation of data on MuSC frequency in geriatric control mice using CD200 and CD63 in the final figure.

(2) Can the authors identify any unique characteristics of Pax7-VCAM-1 GER1-MuSCs using only the data generated in this study, without relying on public databases? For example, reduced expression of Vcam1 and Pax7. The results of such analyses should be presented.

(3) In the senolysis experiment, the authors state that GER1-MuSCs were depleted. However, no data are provided to support this conclusion. Quantitative cell count data would directly address this concern. In addition, the FACS profile corresponding to Figure 4D should be included.

(4) Figure S4: It remains unclear whether DHT enhances regenerative ability through restoration of the VCAM1 expression in GER1-MuSCs, as DHT also acts on non-MuSC populations. Analyses of the regenerative ability of Senolysis+DHT mice may help to clarify this issue.

(5) Why are there so many myonuclear transcripts detected in the single-cell RNA-seq data? Was this dataset actually generated using single-nucleus RNA-seq? This reviewer considers it inappropriate to directly compare scRNA-seq and snRNA-seq results.

Comments on revisions:

Related to Comment#3: The percentage is also influenced by the number of other cell types. Therefore, to demonstrate cell removal, it is necessary to present the absolute number of cells. If the cells were removed and were not replenished from Vcam1+ cells, the absolute number of cells should be reduced.

Related to Comment#4: Without the DHT+Senolysis experiment proposed by this reviewer or related experiments, there is no evidence demonstrating that GERI-MuSCs functionally rejuvenate. The current data only show that VCAM1 expression is restored.

Related to Comment#8: Individual results from 3-4 biological replicates should be shown in Figure 4. It will help readers to recognize the variation of each sample.

<https://doi.org/10.7554/eLife.108680.2.sa3>

Reviewer #2 (Public review):

Kim et al. investigate heterogeneity in aged muscle stem cells using a model that enables lifelong lineage tracing. The questions addressed in the paper are highly relevant to the fields of aging and stem cell biology, and the experimental approach overcomes some of the limitations of previous studies.

The study provides evidence for phenotypic and functional heterogeneity within the lineage-traced aged MuSC pool. However, the data as presented do not yet support the broader conclusions that MuSC abundance is maintained with age or that a previously unrecognized aged MuSC subpopulation has been identified. These claims would require stronger age-matched cohorts, absolute cell counts normalized to tissue mass, and direct comparison to previously described aged muscle stem cell states.

If the core observations were experimentally reinforced, this study could prompt the field to reassess muscle stem cell loss, heterogeneity, and age-associated changes in canonical marker expression in geriatric mice. However, because several of the central claims depend on analyses that are currently incomplete, the conceptual impact should be treated as provisional. The deposited bulk RNA-seq and scRNA-seq datasets should be useful for mapping these states to existing atlases and for re-analysis by groups interested in quiescent and senescent programs in geriatric muscle stem cells.

<https://doi.org/10.7554/eLife.108680.2.sa2>

Reviewer #3 (Public review):

Summary:

The manuscript by Kim et al. describes a MuSC subpopulation that loses VCam expression in geriatric muscle and shows reduced ability to contribute to muscle regeneration. They propose that this population underlies the reported decline of MuSCs in aged mice, suggesting that these cells remain present in geriatric muscle but are overlooked due to low or absent VCam expression. The identification of a subpopulation that changes with aging would be compelling and of interest to the field.

Strengths:

The authors employ a wide range of assays, from in vitro to in vivo systems, to characterize Vcam-low/negative cells from geriatric muscle. The loss of Vcam appears strong in geriatric mice. They further identify CD63 and CD200 as potential surface markers that remain stable with age, thereby enabling the isolation of MuSCs across different age groups.

Weaknesses:

Some issues remain before establishing whether this population represents a true functional subset or explains the reported decline in MuSC numbers in aged mice. Stronger fate assessment of Vcam-low/negative cells is needed to assess their propensity for cell death and whether this contributes to the conclusions. Comparisons include young, middle-aged, and geriatric mice, but not aged (~24 months) mice, which would help comparisons to previous reports of age-related MuSC decline. The suggestion that the Vcam-low/negative population reflects a senescence-like state remains unclear, as these cells display limited canonical senescence markers, exhibit reversible cell-cycle exit, and yet are reported to be sensitive to senolytic treatment. Validation of CD63 and CD200 as reliable age-independent MuSC markers requires further testing, specifically using the Pax7-YFP tracing model and co-labeling in geriatric mice. Finally, the grouping patterns in some analyses suggest that the Vcam-low/negative fraction may be present in only a subset of geriatric mice, raising the possibility that it reflects health status or pathology rather than a consistent aging-associated phenotype.

<https://doi.org/10.7554/eLife.108680.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

(1) As mentioned above, numerous studies have reported that the number of MuSCs declines with aging. The authors' claim is valid, as Pax7 and Vcam1 were widely used for these observations. However, age-related differences have also been reported even when using these markers (Porpiglia et al., Cell Stem Cell 2022; Liu et al., Cell Rep 2013). (a) When comparing geriatric Vcam1⁺ MuSCs with young MuSCs in this study, did the authors observe any of the previously reported differences? (b) Furthermore, would increasing the sample size in Figure 1 reveal a statistically significant difference? The lack of significance appears to result from variation within the young group. (c) In addition, this reviewer requests the presentation of data on MuSC frequency in geriatric control mice using CD200 and CD63 in the final figure.

(a) When comparing geriatric Vcam1⁺ MuSCs with middle aged MuSCs, we found 1,428 DEGs, where 701 genes were downregulated and 727 genes were upregulated (Fig. S3E). Some of the pathways altered were similar to previously reported differences, such as alterations in the autophagy-lysosome related genes and PI3K-Akt Pathways. However, these alterations did not affect the functional integrity of geriatric Vcam1⁺ MuSCs (Fig. 3 A-F). On the other hand, greater alterations were observed in geriatric Vcam1⁻ MuSCs, accompanied by functional impairment. We have added further elaborations in the manuscript to reflect the comment from the reviewer (pg. 17, lines 369-379).

(b) Thank you for this helpful comment. We understand the reviewer's concern that the variability within the young group may contribute to the absence of statistical significance. We respectfully note that the variance observed in the young cohort could be biologically expected rather than technical noise. Multiple studies have shown that young adult MuSCs display great transcriptional and functional heterogeneity from undergoing post-natal

myogenic maturation (e.g., Biressi et al., 2010; Tierney & Sacco, 2016; Motohashi & Asakura, 2014). This broader heterogeneity naturally increases variance in marker distribution within young samples. We would also like to clarify that our main conclusions are not solely based on differences in the overall proportion of YFP⁺ and Lin[−] cells among age groups. Instead, we also rely on the functional and phenotypic heterogeneity that specifically emerges in geriatric MuSCs.

Although the young group shows greater biological variation, the mean values are relatively similar among the groups. Multiple independent datasets in our study including functional performance and molecular profiles consistently show that the total MuSC frequency does not markedly decline with aging. For these reasons, even if the sample size is increased, we do not expect a change in the overall interpretation of this result. We have revised the Results section to acknowledge the variability observed in the young group and to emphasize that total MuSC frequency is not central to the conclusions of this study (pg. 6, lines 129-134).

(c) MuSC frequency in geriatric control mice using CD200 and CD63 in the final figure are in the figure legend of Fig. 5F (pg. 39, line 825-828).

(2) Can the authors identify any unique characteristics of Pax7-VCAM-1 GERI-MuSCs using only the data generated in this study, without relying on public databases? For example, reduced expression of Vcam1 and Pax7. The results of such analyses should be presented.

In Fig S2C, using the bulk-RNA sequencing data generated in this study, we observe reduced expression of both Pax7 and Vcam1 in Pax7-VCAM-1 GERI-MuSCs population. To better highlight this finding, we have added text in the Results section that explicitly describes the reduced Pax7 expression and Vcam1 loss as distinguishing features of Pax7-VCAM-1 GERI-MuSCs in our dataset (pg. 9, lines 199-200).

(3) In the senolysis experiment, the authors state that GERI-MuSCs were depleted. However, no data are provided to support this conclusion. Quantitative cell count data would directly address this concern. In addition, the FACS profile corresponding to Figure 4D should be included.

In Figure 4D we quantified the frequency of VCAM1 Low YFP positive Lin negative MuSCs after senolysis treatment. This analysis shows a clear trend toward a decrease in the GERI subpopulation, although the difference did not reach conventional statistical significance in this experiment (t test p = 0.0596). We have therefore revised the text to describe this as a reduction trend rather than complete depletion, and we now explicitly report the p value in the results section (pg. 12, line 270-272). Furthermore, representative FACS profiles for Figure 4D is now included with the quantification (pg. 38, line 811-814).

(4) Figure S4: It remains unclear whether DHT enhances regenerative ability through restoration of the VCAM1 expression in GERI-MuSCs, as DHT also acts on non-MuSC populations. Analyses of the regenerative ability of Senolysis+DHT mice may help to clarify this issue.

We thank the reviewer for this important insight. We agree that DHT can act on non-stem cell populations in the muscle environment and therefore we cannot conclusively attribute the improved regenerative performance solely to restoration of VCAM1 expression in GERI-MuSCs. To address this concern, we have revised the discussion to explicitly state this limitation and to clarify that DHT may influence multiple cell types that contribute to muscle regeneration. We also indicate that combined senolysis plus DHT treatment would be an informative future approach, although additional animal experiments were not feasible within the scope of the current study (pg. 18, line 382-390).

(5) *Why are there so many myonuclear transcripts detected in the single-cell RNA-seq data? Was this dataset actually generated using single-nucleus RNA-seq? This reviewer considers it inappropriate to directly compare scRNA-seq and snRNA-seq results.*

Regarding the question of why many myonuclear transcripts were detected and whether this dataset was generated using single nucleus RNA sequencing, we confirm that the experiments were performed using single cell RNA sequencing. The presence of myonuclear transcripts likely reflects partial nuclear leakage or fragmentation during the enzymatic dissociation of aged muscle tissue. This is a known technical issue when preparing single cell suspensions from adult or geriatric skeletal muscle.

To avoid inappropriate interpretation, we identified the myonuclear transcript enriched cluster and excluded it from all downstream analyses that involve MuSC comparison. Therefore, our major conclusions do not rely on this cluster. We have revised the Results text to clearly state that the dataset was generated using single cell RNA sequencing and to explain how myonuclear transcript-positive cells were handled (pg. 8, lines 176-181).

Reviewer #2 (Public review):

In this study, Kim et al. explore the heterogeneity within the aged MuSC population using a mouse model that enables lineage tracing of MuSCs throughout life. The questions addressed in the manuscript are highly relevant to the fields of aging and stem cell biology, and the experimental approach overcomes limitations of earlier studies. However, some of the claims would benefit from additional data analysis, and the central claim of the identification of a "previously unrecognized subpopulation" of aged MuSCs should be evaluated in light of prior work that has also examined MuSC heterogeneity in aging.

Specific points:

(1) As a general comment that is transversal to multiple figures, several experiments should include a direct comparison to a young cohort. Previous studies have shown that the depletion of subpopulations with aging is observed early in the aging process, for example, the loss of Pax7-high MuSCs is observed already in 18-month-old mice (Li, 2019, doi: 10.15252/embj.2019102154). Using only mice at 12-14 months as the control group is therefore insufficient to claim that no changes occur with aging.

We thank the reviewer's suggestion for comparing the aged mice to a young cohort and we acknowledge that previous studies have observed depletion of subpopulations is observed early in the aging process. However, this study is specifically designed to delineate the transition from middle aged to geriatric stages, rather than to characterize differences that are already well established in young versus geriatric comparisons. Previous studies have extensively documented the decline in MuSC function between young and aged animals, whereas the process and timing by which these changes emerge remain unclear. Our results show that major alterations in MuSC phenotype and identity are detected predominantly in the geriatric stage rather than at the middle aged stage. To avoid any misunderstanding, we have revised the text to clearly state that the primary objective of this work is to define the critical shift that occurs from middle aged to geriatric muscle stem cells (page 3-4, line 67-71).

(2) One of the central claims of the manuscript is a challenge to the notion that MuSCs number declines with age. However, the data analysis associated with the quantification of YFP+ cells needs to be expanded to support this conclusion. The authors present YFP+ cells only as a proportion of Lin-neg cells. Since FAP numbers are known to decrease with aging, a stable proportion of YFP+ cells would simply indicate that MuSCs decline at the same rate as FAPs. To more accurately assess changes in MuSC abundance, the authors

should report absolute numbers of YFP+ cells normalized to tissue mass (cells/ mg of muscle).

We thank the reviewer for this helpful suggestion. We agree that a proportion based analysis alone does not fully exclude the possibility that MuSCs and FAPs decrease at similar rates during aging. At the time of isolation, muscle mass was not recorded, so we are unable to report YFP⁺ cell numbers normalized to tissue weight as requested. To partially address this limitation, we have now clarified our gating strategy in the methods and Figure 1 to explicitly indicate Sca1⁺ FAP exclusion (pg. 6, line 121-122, pg. 22, lines 460-463). These analyses do not support a major selective loss of MuSCs relative to other mesenchymal populations with aging.

(3) The authors emphasize that several studies use VCAM1 as a surface marker to identify MuSCs. However, many other groups rely on α7-integrin, and according to Figure 1D, the decline in ITGA7 expression within the YFP+ population is not significant. Therefore, the suggestion that MuSC numbers have been misquantified with aging would apply only to a subset of studies. If the authors can demonstrate that YFP+ cell numbers (normalized per milligram of tissue) remain unchanged in geriatric mice, the discussion should directly address the discrepancies with studies that quantify MuSCs using the Lin-α7-integrin+ strategy.

We thank the reviewer for this important comment. We agree that VCAM1 is only one of several commonly used surface markers for MuSC identification and that many studies quantify MuSCs using the Lin negative and ITGA7 positive strategy. That is why in our study, in addition to VCAM1, we also examined ITGA7 expression within the YFP positive population. Although the mean ITGA7 level did not significantly decline, the variance among geriatric MuSCs was significantly increased based on the F test. This supports the idea that aging does not uniformly reduce marker expression but instead increases phenotypic instability, which could lead to under detection of a subset of MuSCs even when ITGA7 is used as the primary marker. We have added this interpretation to the Discussion (pg. 16, lines 346-355).

(4) The authors focus their attention on a population of VCAM-low/VCAM-neg subpopulation of MuSCs that is enriched in aging. However, the functional properties of this same population in middle-aged (or young) mice are not addressed. Thus, it remains unclear whether geriatric VCAM-low/VCAM-neg MuSCs lose regenerative potential or whether this subpopulation inherently possesses low regenerative capacity and simply expands during aging.

We thank the reviewer for this comment. In young and middle aged mice, the VCAM low or VCAM negative population is extremely small, nearly absent in most samples. The emergence and expansion of this population is therefore a feature that becomes detectable only at the geriatric stage. Given that these cells are not present in appreciable numbers earlier in life, the reduced regenerative performance observed in geriatric VCAM1^{low} MuSCs likely reflects a phenotype that arises during aging rather than an inherent property of a pre-existing subpopulation. We have added this clarification to the Results section (pg. 7, lines 142-146).

(5) According to Figure 1F, the majority of MuSCs appear to fall within the category of VCAM-low or VCAM-neg (over 80% by visual estimate). It would be important to have an exact quantification of these data. As a result, the assays testing the proliferative and regenerative capacity of VCAM-low/negative cells are effectively assessing the performance of more than 80% of geriatric MuSCs, which unsurprisingly show reduced efficiency. Perhaps more interesting is the fact that a population of VCAM-high geriatric MuSCs retains full regenerative potential. However, the existence of MuSCs that preserve regenerative potential into old age has been reported in other studies (Garcia-Prat, 2020, doi: 10.1038/s41556-020-00593-7; Li, 2019, doi: 10.15252/embj.2019102154). At this

point, the central question is whether the authors are describing the same aging-resistant subpopulations of MuSCs using a new marker (VCAM) or whether this study truly identifies a new subpopulation of MuSCs. The authors should directly compare the YFP+VCAM+ aged cells with other subpopulations that maintain regenerative potential in aging.

We thank the reviewer for this comment. First, in response to the request for precise quantification, we now provide the proportions of VCAM1-high and VCAM1-low/negative MuSCs in each age group in the figure legends for Fig.1F (pg. 34-35, lines 765-772). In geriatric mice, VCAM1 low/negative MuSCs represent approximately $44.6\% \pm 35.7\%$, whereas VCAM-high MuSCs represent $3.9\% \pm 1.8\%$. The substantial variability reflects mouse-to-mouse heterogeneity at very advanced ages.

Importantly, our conclusions do not rely solely on the observation that a large fraction of geriatric MuSCs exhibit reduced regenerative potential. Rather, the VCAM-low state represents a transcriptionally and functionally distinct subpopulation that emerges specifically in the geriatric stage, and exhibits molecular signatures not present in young or mid-aged MuSCs. We have expanded the Results and Discussion to clarify this point.

Regarding whether VCAM-high geriatric MuSCs correspond to previously reported “aging-resistant” MuSCs (e.g., Garcia-Prat 2020; Li 2019), we agree that there may be conceptual overlap, as both populations retain regenerative activity. However, those studies identified resilient MuSCs based on mitochondrial or Pax7-high properties, whereas our classification is based on surface VCAM1 intensity, and we currently lack direct evidence that these populations are equivalent. We have therefore added a statement acknowledging this possibility while clarifying that our work does not claim that VCAM1-high MuSCs represent a newly discovered resilient subset, but instead focuses on the emergence and characterization of the VCAM-low dysfunctional subpopulation (pg. 16, lines 346-355).

(6) In Figure 3F, it is unclear from the data presentation and figure legend whether the authors are considering the average of fiber sizes in each mouse as a replicate (with three data points per condition), or applied statistical analysis directly to all individual fiber measurements. The very low p-values with n=3 are surprising. It is important to account for the fact that observations from the same mouse are correlated (shared microenvironment, mouse-specific effects) and therefore cannot be considered independent.

We thank the reviewer for raising this important statistical point. We fully agree that individual myofibers from the same mouse are not independent biological replicates. In morphometric analyses of regenerated muscle, however, it is standard practice to analyze the full CSA distribution across all regenerated fibers, as the distribution itself (rather than a per-mouse mean) provides the biologically relevant measure of regeneration quality.

The original analysis therefore treated each regenerated fiber as a component of the overall CSA distribution, not as an independent biological replicate, and the statistical comparison was performed at the level of distributions rather than per-mouse replication. We agree that per-mouse averaged CSA values would also be informative, but the raw data were not archived in a format that allows reconstruction of mouse-specific fiber subsets.

Importantly, the group-level CSA distribution differences are robust and remain clearly detectable regardless of statistical approach. We have added clarification in the figure legend to explicitly describe how CSA measurements were obtained and analyzed mouse (pg. 36, lines 796-800).

(7) Regarding Figure 5, it is unclear why ITGA7, a classical surface marker for MuSCs that appears unchanged in aged YFP+ MuSCs (Fig. 1F), is considered inadequate for detecting and isolating GERI-MuSCs.

We thank the reviewer for raising this point. As shown in Figure 1F, the mean ITGA7 expression level does not significantly decline in geriatric YFP positive MuSCs. However, the variance of ITGA7 expression is significantly increased in geriatric MuSCs based on the F test, indicating instability in surface marker expression. This suggests that a fraction of MuSCs may fall below the conventional gating threshold for ITGA7 during aging. Therefore, ITGA7 remains effective for identifying a large portion of MuSCs but may under detect the subset of geriatric MuSCs with reduced marker expression. We have revised the Discussion to clarify this point (pg. 16, lines 346-355).

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) Figure 3B: In the colony formation assay, the authors should specify the number of biological replicates and the number of cells analyzed per mouse.

We have now added the number of biological replicates and the number of cells analyzed per mouse in the figure legend of Figure 3B (pg. 37, lines 790-791).

(2) Figure 3F: The replication number is indicated as $n = 3$, which appears to refer to the number of transplanted mice. How many myofibers were analyzed in each transplanted mouse? The authors should provide a more detailed description of the methodology in the Figure legend or M&M.

We thank the reviewer for the question and clarify that $n = 3$ refers to three independent transplanted mice per group. For each mouse, the entire TA muscle was cryosectioned and immunostained, and all regenerated fibers containing centrally located nuclei were included in the CSA quantification. We have added clarification in the Figure legend to indicate that quantification was performed on all regenerated fibers from each mouse (pg. 37, lines 796-800).

(3) Figure 4: The RNA-seq results are presented as a single dataset per sample. If multiple experiments were performed, individual datasets should be shown. Replicated analyses are essential to ensure the reliability of the findings.

In response to the reviewer comment, we confirm that the RNA sequencing in Figure 4 was performed with 3-4 independent biological replicates for each condition. These replicates showed very consistent sequencing quality and gene expression profiles and were therefore combined for the differential expression analysis. We have revised the materials and methods to clearly describe the number of biological replicates and the analysis workflow. (pg. 25, lines 543).

(4) Line 148: If the authors examined MyoG expression, it should be described as committed myoblasts.

We have now changed the term from myoblasts to committed myoblasts (pg. 8, line 168).

(5) Typo and Referencing Errors:

(a) Line 244: The term 'Antide' appears to be a typo.

We thank the reviewer for noting this point. ‘Antide’ is not a typo but the correct name of a GnRH antagonist (Antide acetate). To avoid confusion, we have revised the text to specify ‘Antide, a GnRH antagonist’ at its first mention (pg. 13, line 289).

(b) Lines 278, 280: Please correct Figure 5H to Figure 5F.

We apologize for this error. We have fixed the figure notations accordingly (pg. 15, lines 326-330).

(c) Some references are incomplete or inappropriate (ex. line 49, line 71, line 86, line 109).

We apologize for this error. We have fixed the references accordingly (pg. 4, line 94, pg.6, line 117).

(d) Line 49: Skeletal muscle regeneration is orchestrated primarily by tissue resident stem cells, known as muscle stem cells (MuSCs) or satellite cells (Rella et al., 2021). The following paper should be cited:

Satellite cell of skeletal muscle fibers.

MAURO A. *J Biophys Biochem Cytol.* 1961 Feb;9(2):493-5.

The reference has been revised (pg. 3, line 49).

(e) Line 109: Paired box protein 7 (Pax7) is a transcription factor widely recognized as a defining marker of MuSCs (Sambasivan et al., 2011). The following paper should be cited:

Pax7 is required for the specification of myogenic satellite cells.

Seale P, Sabourin LA, Girgis-Gabardo A, Mansouri A, Gruss P, Rudnicki MA. *Cell.* 2000 Sep 15;102(6):777-86.

The reference has been revised (pg.6, line 117).

(6) Lines 73-74: Many rejuvenation studies define 'aged' mice as 12 to 24 months old. This reviewer is not aware of any studies that have examined 12-month-old MuSCs as a model of aging.

We apologize for this error. We have fixed the numbers to 18 months accordingly (pg. 4, line 94).

Reviewer #3 (Recommendations for the authors):

(1) Geriatric versus aged mice in the MuSC subpopulation analysis. The authors use geriatric mice (>28 months) to demonstrate the loss of VCam expression in MuSCs and propose that this accounts for previous reports of decreased MuSC numbers in aged contexts. However, as noted in their introduction, most reports use "aged" mice, which are typically around 24 months old, which is biologically distinct from the geriatric stage. This distinction makes it difficult to conclude that the reported decline in MuSC numbers in aged mice can be explained by the phenomenon observed only in geriatric mice (Line 289). The authors should test whether VCam expression is altered in aged (24-month-old) mice to strengthen this argument.

We appreciate the reviewer’s thoughtful comment and agree that 24 month old mice are commonly used as an aged reference in the literature. However, prior studies using 18 to 24 month old animals have reported inconsistent results regarding whether and to what extent MuSCs decline during this period. To avoid ambiguity from intermediate aging stages, we

purposefully selected geriatric mice older than 28 months, a condition under which MuSC depletion has been more consistently reported in previous studies. Notably, our data show that even at this stage MuSC abundance is not dramatically reduced, which makes it unlikely that a robust decline would already be present at 24 months. We have clarified this rationale in the revised text. Although investigating the precise timing of the emergence of these changes at earlier time points is an important future direction, it is beyond the scope of the present study.

(2) Variability and bimodal distributions.

Figure 1b: The decline in VCAM+ MuSCs in geriatric mice shows high variability - 3 of 7 replicates align more closely with young/mid-aged levels. Please clarify this variability.

We thank the reviewer for pointing out the variability. We agree that there is heterogeneity in the extent of VCAM1 reduction across geriatric mice. This variability likely reflects animal-to-animal differences in the onset and progression of aging-related phenotypes, which are known to vary at very advanced ages. Importantly, despite this variability, all geriatric samples contain a detectable VCAM1 low population that is not observed in young or middle-aged mice, and the overall trend is consistent across all replicates. We have clarified this in the revised manuscript (pg. 6, lines 125-127).

Figure 1c: While the Mid and Geriatric groups are tightly clustered, the Young group appears bimodal, which challenges the claim (Line 118) that values are "comparable across ages." Since all males were used and it is not sex related, what is driving this bimodal distribution?

We appreciate the reviewer's observation regarding the variability in the young group. Muscle stem cells in young adult mice are known to encompass diverse transcriptional and functional substates, which contribute to greater biological heterogeneity at this stage (Biresi et al. 2010; Tierney & Sacco 2016; Motohashi & Asakura 2014). As aging progresses, these substates gradually converge toward a common functional phenotype, resulting in more uniform profiles in middle-aged and geriatric mice. Therefore the bimodal appearance in the young group likely reflects the broader developmental heterogeneity of early adult MuSCs rather than a technical discrepancy. We have added this explanation to the revised in the results section (pg.6. lines 129-134).

Figure 4D: Geriatric replicates also display a trimodal distribution. This should be addressed throughout - what is causing these types of distribution, and how does this impact significance tests and conclusions?

We appreciate the reviewer's observation regarding the multimodal distribution. We interpret this pattern as reflecting increased individual variability that becomes more pronounced at the geriatric stage. Even though aging affects all mice, the extent and timing of age-related phenotypic changes can vary considerably across individuals at very advanced ages. This leads to broader divergence in VCAM1 expression states among geriatric mice. Therefore, when we look at the correlation between VCAM1 High and VCAM1 Low/- population, there exists a significant negative correlation between the two populations (Fig. S3F). We have clarified this interpretation in the text and note that the statistical analysis was performed using the mouse as the biological replicate, so this variability does not alter the overall conclusion (pg.12-13, lines 270-278).

(3) The fate of the Vcam-low/negative cells should be better assessed. For example, Line 180: Colony formation is low/absent in VCAM-low/- cells. Are these cells still viable? Cell death assays are needed. Is expansion capacity truly impaired, or are the cells simply non-viable? Using gene expression as the only means (Line 300) to suggest not dying is insufficient.

We thank the reviewer for this important point. As per the reviewer's analysis, there is lack of direct evidence to show that these cells are viable and apoptosis or viability assay would further strengthen our research. However, we carefully suggest that they are viable from the fact that these cells can be isolated by FACS and generate high quality RNA sequencing libraries, which would not be possible if they were undergoing cell death. Moreover, the transcriptomic data indicate upregulation of stress response and senescence associated pathways rather than apoptotic or necrotic signatures. These findings suggest that VCAM low or negative cells are alive but exhibit reduced proliferative and regenerative capacity. We have revised the text to clarify that our data reflect impaired function rather than loss of viability and that apoptosis assays represent a direction for future investigation (pg. 16, 360-366).

(4) Transplant assays are suggestive, but could use additional characterization. Lines 191 & Figure 3E-F: While representative images match quantification, areas at the edge of VCAM-low/- TAs show signs of regeneration. Please include lower-magnification images. Additionally, assess early post-transplant engraftment efficiency - do certain populations experience a higher loss rate (cell death)? YFP-tracing would also help confirm the donor contribution to fibers.

While we did not collect additional early time-point samples for new engraftment analyses, we carefully re-examined all available transplantation data, including the distribution and density of YFP⁺ donor-derived cells in early post-injury sections. We did not observe patterns suggestive of differential early cell loss between VCAM-high and VCAM-low groups. Thus, although we cannot formally quantify early engraftment efficiency, the existing evidence does not support a model in which differential donor-cell retention accounts for the observed regenerative differences.

Also, we attempted direct YFP co-staining of regenerated myofibers, but as reported by several groups, YFP signal within mature or regenerating myofibers is often diminished or inconsistent after fixation and permeabilization, making reliable fiber-level YFP detection technically challenging in our system. Therefore, instead, we confirmed donor contribution using PBS-injected control muscles, which lack donor MuSCs, and showed that PBS-injected muscles never generated YFP⁺ fibers. This demonstrates that endogenous MuSCs do not contribute to YFP⁺ myofibers in our model, and therefore indirectly supports our suggestion that any YFP⁺-regenerated fiber necessarily originates from transplanted donor cells. We hope the reviewer understands the technical limitations.

(5) Figure S3D: mRNA profiling suggests Mid-aged MuSCs are more distinct from Geriatric Vcam-hi than expected. This should be addressed or at least elaborated on in text.

We appreciate this insightful comment. We agree that mid aged VCAM high MuSCs show detectable transcriptional differences from geriatric VCAM high cells. This pattern likely reflects the fact that some aging related molecular changes begin to accumulate gradually during the middle aged stage even before overt functional decline or VCAM1 loss becomes evident. Importantly, however, these transcriptomic shifts do not lead to the emergence of the VCAM low dysfunctional phenotype that is uniquely present in geriatric muscle. We have added clarification to the text noting that molecular alterations arise progressively while the major phenotypic transition in VCAM1 expression and regenerative impairment occurs at the geriatric stage (pg.11, 238-244).

(6) The conclusion of senescence needs more support. Lines 218-226: p16 is elevated in VCAM-low/- cells, but drawing conclusions on senescence from 1-2 markers (mRNA) is insufficient. DQ Treatment: It's unclear how DQ alters cell composition in the absence of clear senescence markers (besides p16). Since DQ targets BCL-2/anti-apoptotic pathways,

analyzing these signaling cascades is necessary. Line 255: The term "terminally senescent" is contradictory. These may be pre-senescent. It's also surprising DQ would target such cells, and further clarification is needed. Lines 307-313: Proposing a revised definition of senescence is premature. These cells may be pre-senescent, and multiple ways to senescence exist (replicative, stress-induced, etc.). Please clarify.

We agree with the reviewer that the term 'terminally senescent' may be premature and potentially contradictory. Although p16 is elevated in this population, we acknowledge that one or two mRNA markers are insufficient to establish bona fide senescence, and that multiple senescence programs exist, including replicative, stress-induced, and mitochondrial-associated pathways. We have revised this to 'senescent-like' throughout the manuscript to better reflect the complexity of this state. Also, although beyond the scope of this study, we now emphasize that future studies incorporating additional senescence markers, functional assays, and lineage tracing will be required to determine the precise senescence status of VCAM-low MuSCs (pg.17-18, lines 381-392).

Regarding DQ treatment, we agree that DQ is not selective for senescent cells, as it targets BCL-2-related survival pathways. The reduction of VCAM-low cells after DQ treatment therefore indicates increased dependence on survival signaling in this population rather than providing direct evidence of senescence. We have revised the text to clarify this interpretation (pg.12-13, lines 270-278).

(7) Figure 5C: The Pax7+ cells appear interstitial rather than sublaminal. This raises questions about the specificity of staining. Providing lower-magnification images with these as insets may help.

We thank the reviewer for this helpful comment. We agree that the high-magnification image in Figure 5C may give the impression that Pax7⁺ cells are interstitial due to the limited field of view. We regret to inform the reviewer that low-magnification images for this sample are not available as these images were obtained via confocal imaging where we only recorded areas of interest. Therefore, we are unable to provide an additional panel at this time and we hope the reviewer understand.

(8) CD63 and CD200 expression on Pax7-YFP traced cells. Figure 5: YFP-traced geriatric MuSCs co-stained for CD63 and CD200 are essential. Current data only show expression in Young traced cells. It's crucial to confirm whether protein/surface expression persists in geriatric YFP+ (traced) cells. The current Figure 5 F does not appear to include YFP tracing for geriatrics.

We thank the reviewer for highlighting the importance of confirming CD63 and CD200 expression specifically in Pax7-YFP traced MuSCs from geriatric muscle. The datasets shown in Figure 5F were generated from wild-type C57BL/6 mice using a standard MuSC gating strategy rather than Pax7-YFP animals. All geriatric Pax7-YFP mice available for this study were exhausted during earlier experiments, and additional tissue is not available for new co-staining or FACS analyses. We now state this technical limitation in the manuscript and clarify that the geriatric CD63/CD200 data were obtained from conventionally isolated MuSC populations rather than YFP-traced cells (pg.18-19, lines 407-416).

Minor points:

(1) Please show the outliers in addition to the concentric circles. Figures 1B, C, and F are examples, but this should be addressed throughout.

Outliers have been added where applicable.

(2) Figure 2C: Was a significance test performed between the 5 dpi and "geri" fractions?

We thank the reviewer for this important point. We have now performed the requested statistical comparison between the 5 dpi fraction and the geriatric VCAM1-defined subpopulations using the same analysis framework applied in Figure 2 (Kruskal–Wallis test followed by Dunn’s multiple comparisons).

While 5 dpi MuSCs differed significantly from young MuSCs (adjusted $p = 0.0139$), the comparisons between 5 dpi and each geriatric subgroup (VCAM-high, -mid, and -low) did not reach statistical significance after correction for multiple testing (adjusted $p = 0.17$, 0.15 , and 0.17 , respectively). These results have been added to the revised Figure 2C corresponding figure legend (pg. 36, lines 777-780).

Importantly, we now clarify in the text that although 5 dpi muscles display a prominent increase in VCAM1-high cells at the population level, this increase does not statistically exceed the variability observed within geriatric subpopulations under the conservative non-parametric testing framework used.

(3) Line 155: The phrase "Surprisingly, all clusters mapped to quiescent clusters" is misleading; this is expected given the population type.

We thank the reviewer for this helpful comment. We have revised the sentence to remove the misleading wording and now describe the observation more accurately (pg. 8 lines 180-181).

(4) Line 211: The figure notation should be corrected from Figure S4E to Figure S3E.

We apologize for this error. We have fixed the figure notation for Figure S4E to S3E (pg. 11, line 247).

(5) Line 216: "All of which" seems overstated. Many populations share similar profiles with minor differences.

We appreciate the reviewer’s comment. We agree that the phrase “all of which” overstated the degree of divergence among clusters. We have revised the wording to more accurately reflect the data (pg. 11-12, lines 252-253).

(6) Line 270: The notations for panels D, E, and F need to be updated to match the figure. Panel "H" is not indicated in Figure 5.

We apologize for this error. We have fixed the figure notations accordingly (pg. 15, lines 326-336).

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